

# **User's Guide**

## **QConvergeConsole CLI**

2400/2500/3200/8100/8200 Series

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# Preface

This guide describes the QConvergeConsole<sup>®</sup> CLI, which is used to configure and manage QLogic Fibre Channel adapters, Intelligent Ethernet adapters, and Converged Network Adapters within SANs.

## What's in this Guide

This guide contains the basic information you need to get started with the QConvergeConsole CLI.

This preface explains the QConvergeConsole CLI help system, typographic conventions used in this guide, lists related documents, specifies the intended audience, refers you to the QLogic license agreements, and provides technical support and contact information.

The remainder of the user's guide contains the following sections:

- [Section 1, System Requirements](#) lists the hardware, software requirements, and operating system requirements needed for the successful operation of the QConvergeConsole CLI.
- [Section 2, Installing and Uninstalling QConvergeConsole CLI](#) describes how to install and uninstall.
- [Section 3, Getting Started](#) describes how to start the QConvergeConsole CLI on Windows and Linux platforms. This section also describes how to display detailed information about command parameters and options.
- [Section 4, NIC Noninteractive Commands](#) describes the noninteractive NIC command syntax and parameters.
- [Section 5, Fibre Channel Noninteractive Commands](#) describes the noninteractive Fibre Channel command syntax and parameters.
- [Section 6, iSCSI Noninteractive Commands](#) describes the noninteractive iSCSI command syntax and parameters.

## What's in the Help System

The QConvergeConsole CLI help system (`qauc li - h`) contains a condensed version of the NIC, Fibre Channel, and iSCSI noninteractive commands described in Sections [4](#) through [6](#).

## Intended Audience

This guide is for system administrators who are responsible for installing, configuring, and managing QLogic adapters using the QConvergeConsole CLI.

## Related Materials

For additional help installing or using the QConvergeConsole, refer to the following related documents posted on the QLogic Web site  
<http://driverdownloads.qlogic.com>:

*QConvergeConsole CLI Readme*

*QConvergeConsole CLI Release Notes*

For information about the QConvergeConsole GUI Utility, refer to the following document:

*QConvergeConsole User's Guide*

## Documentation Conventions

This guide uses the following documentation conventions:

- **NOTE:** provides additional information.
- **CAUTION!** indicates the presence of a hazard that has the potential of causing damage to data or equipment.
- **WARNING!!** indicates the presence of a hazard that has the potential of causing personal injury.
- Text in **blue** font indicates a hyperlink (jump) to a figure, table, or section in this guide, and links to Web sites are shown in **underlined blue**. For example:
  - ❑ **Table 9-2** lists problems related to the user interface and remote agent.
  - ❑ See **"Installation Checklist"** on page 3-6.
  - ❑ For more information, visit **www.qlogic.com**.
- Text in **bold** font indicates user interface elements such as a menu items, buttons, check boxes, or column headings. For example:
  - ❑ Click the **Start** button, point to **Programs**, point to **Accessories**, and then click **Command Prompt**.
  - ❑ Under **Notification Options**, select the **Warning Alarms** check box.

- Text in `Courier` font indicates a file name, directory path, or command line text. For example:
  - ❑ To return to the root directory from anywhere in the file structure:  
Type `cd /root` and press **ENTER**.
  - ❑ Enter the following command: `sh ./install.bin`
- Key names and key strokes are indicated with UPPERCASE:
  - ❑ Press CTRL+P.
  - ❑ Press the UP ARROW key.
- Text in *italics* indicates terms, emphasis, variables, or document titles. For example:
  - ❑ For a complete listing of license agreements, refer to the *QLogic Software End User License Agreement*.
  - ❑ What are *shortcut keys*?
  - ❑ To enter the date type *mm/dd/yyyy* (where *mm* is the month, *dd* is the day, and *yyyy* is the year).
- Topic titles between quotation marks identify related topics either within this manual or in the online help, which is also referred to as *the help system* throughout this document.

## License Agreements

Refer to the *QLogic Software End User License Agreement* for a complete listing of all license agreements affecting this product.

## Technical Support

Customers should contact their authorized maintenance provider for technical support of their QLogic products. QLogic-direct customers may contact QLogic Technical Support; others will be redirected to their authorized maintenance provider.

Visit the QLogic support Web site listed in [Contact Information](#) for the latest firmware and software updates.

## Training

QLogic offers training for technical professionals for storage networking, converged networking, and High Performance Computing (HPC) networking products. From the main QLogic web page at [www.qlogic.com](http://www.qlogic.com), click the **Support** tab at the top, and then click the **Training and Certification** on the left. The QLogic Global Training Portal offers online courses, certification exams, and scheduling of in-person training.

Technical Certification courses include installation, maintenance and troubleshooting QLogic products. Upon demonstrating knowledge using live equipment, QLogic awards a certificate identifying the student as a Certified Professional. You can reach the training professionals at QLogic by e-mail at [training@qlogic.com](mailto:training@qlogic.com).

## Contact Information

Please feel free to contact your QLogic approved reseller or QLogic Technical Support at any phase of integration for assistance. QLogic Technical Support can be reached by the following methods:

**Web site:** <http://support.qlogic.com>

**E-mail:** [support.qlogic.com](mailto:support.qlogic.com)

QLogic Technical Support for products under warranty is available during local standard working hours excluding QLogic observed holidays.



# 1 System Requirements

The QConvergeConsole CLI is an management utility that centralizes management and configuration of QLogic adapters within the entire network (LAN and SAN). For optimum performance, the QConvergeConsole CLI requires the hardware, software, and operating systems listed in this section:

- [Hardware Requirements](#)
- [Software Requirements](#)
- [Operating System Requirements](#)
- [Supported QLogic Adapters](#)

## Hardware Requirements

The minimum hardware requirements for the workstation server are as follows:

- **Server.** Single-processor or multiprocessor server or workstation. See [“Operating System Requirements” on page 1-2](#) for a list of operating systems.  
Pentium<sup>®</sup> II processor, 300 MHz
- **Memory.** 64MB of physical RAM to run the QConvergeConsole CLI
- **Hard disk space.** About 26MB disk space.

The minimum hardware requirements for the adapter are as follows:

- **Adapters.** One or more of the QLogic adapters listed under [“Supported QLogic Adapters” on page 1-3](#).

The minimum requirements for a SAN environment are as follows:

- **Storage.** Fibre Channel or iSCSI devices, such as disks and RAID subsystems. The QConvergeConsole CLI supports most Fibre Channel and iSCSI devices. See the *QLogic Storage Networking Interoperability Guide*, which you can download from the following QLogic Web page:  
<http://connect.qlogic.com/interopguide/info.asp>

**NOTE:**

Tape devices are shown as part of the configuration, but are not fully supported. Only persistent binding is supported.

## Software Requirements

The software requirements for the server on which your adapter is physically installed are as follows:

- QLogic adapter drivers for your operating system

**NOTE:**

Refer to the QLogic Web site for QLogic adapter drivers and agents:  
<http://driverdownloads.qlogic.com>.

- TCP/IP protocol for Windows remote management
- Administrative privileges to perform management functions

## Operating System Requirements

The QConvergeConsole CLI runs on the following platforms:

- Windows® Server® 2003, Windows Server 2008
- Red Hat® Linux® Advance Server/Enterprise Server
- Novell SUSE® Linux Enterprise Server (SLES)

**NOTE:**

For the supported operating system versions, refer to the Readme where the QConvergeConsole CLI is posted on the QLogic Web site:  
<http://driverdownloads.qlogic.com>.

## Supported QLogic Adapters

The supported QLogic-branded adapters are as follows. For support of OEM-branded adapters, please contact your OEM.

- 2400 and 2500 Series Fibre Channel Adapters
- 3200 Series Intelligent Ethernet Adapters
- 8100 and 8200 Series Converged Network Adapters

**NOTE:**

For information about QLogic adapters, refer to the QLogic Web site at <http://www.qlogic.com/Products/adapters>.



# 2

## Installing and Uninstalling QConvergeConsole CLI

This section describes how to download, install, and uninstall the QConvergeConsole CLI.

- [Downloading the QConvergeConsole CLI](#)
- [Installing the QConvergeConsole CLI](#)
- [Uninstalling the QConvergeConsole CLI](#)

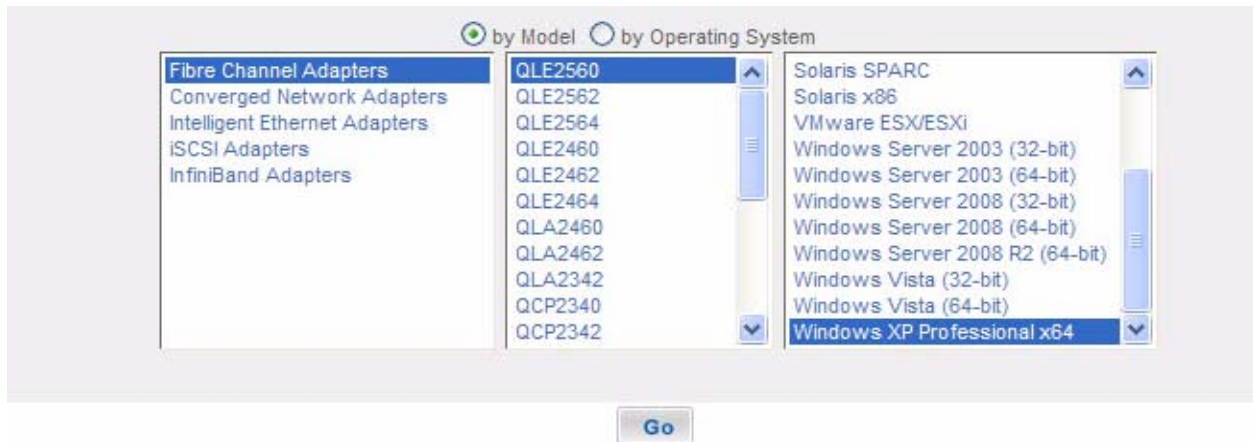
### **NOTE:**

The QConvergeConsole CLI is designed to configure QLogic adapters on the local machine (where it is installed). It cannot configure adapters on remote machines.

## Downloading the QConvergeConsole CLI

To download the QConvergeConsole CLI from the QLogic Web site:

1. On the QLogic home page (<http://www.qlogic.com>), click the **Downloads** tab, then select **QLogic Products**.
2. In the selection box ([Figure 2-1](#)), click **Fibre Channel Adapters**, **Intelligent Ethernet Adapters**, or **Converged Network Adapters** in the left column, your adapter model number in the middle column, and your operating system in the right column.



**Figure 2-1. Download Selections (Example)**

3. Click **Go**.
4. Scroll down to the **Management Tools** table and click the appropriate version of the QConvergeConsole CLI for your operating system.
5. On the End User Software License Agreement page, read this important information, and then click **I Agree** to continue.
6. On the File Download dialog box, click **Save**. Select a directory on your system, and then click **Save As** to download the file. The package file has one of the following formats, depending on the operating system:
  - Windows:  
QConvergeConsoleCLI-AA.BB.CC-DD.EE\_win\_<Subtype>.msi
  - Linux:  
QConvergeConsoleCLI-AA.BB.CC-DD.i386.rpm
7. In the **Management Tools** table, select and download the Readme and Release Notes files for the appropriate version of the QConvergeConsole CLI.

## Installing the QConvergeConsole CLI

The installation procedures differ depending on the operating system:

- [Installing the QConvergeConsole CLI in a Windows Environment](#)
- [Installing the QConvergeConsole CLI in a Linux Environment](#)

### Installing the QConvergeConsole CLI in a Windows Environment

You can install the QConvergeConsole CLI from the command prompt using the Microsoft® Windows Installer (MSI). Use one of the following methods:

- [Standard Windows Interactive \(CLI\) Installation](#)
- [Quiet or Unattended Windows Installation](#)
- [Passive Windows Installation](#)
- [Overwrite Previous Windows Installations](#)

#### **NOTE:**

You can also configure the installation using MSI commands. To see a summary of MSI commands, at the command prompt, type: **msiexec**. To obtain more information about MSI, visit the Microsoft Web site.

#### Standard Windows Interactive (CLI) Installation

To begin a standard installation of the QConvergeConsole CLI on a Microsoft Windows operating system, at a command line type one of the following:

```
QCCCLI.msi  
msiexec /i QCCCLI.msi
```

The default directory for the QConvergeConsole CLI application is:

```
Program files\QLogic Corporation\QCCCLI
```

If you want a different directory, type it in the command line. For example:

```
QCCCLI.msi installldir="directory"
```

The preceding command installs the QConvergeConsole CLI in the directory named **directory**. Use full path names.

## Quiet or Unattended Windows Installation

Type the following for a quiet (silent) installation using default values. For example, the following command installs silently using defaults and does not show any errors:

```
QCCCLI.msi /q
```

## Passive Windows Installation

Type the following for a passive installation using default values. For example:

```
QCCCLI.msi /passive
```

The preceding command installs with minimum interaction, showing only the progress bar and any errors.

## Overwrite Previous Windows Installations

Type the following for an installation that overwrites any previous installations *without* asking for confirmation. Type:

```
QCCCLI.msi /i forceinstall=true
```

## Installing the QConvergeConsole CLI in a Linux Environment

To install the QConvergeConsole CLI on a Linux platform, type the following from the command form:

```
rpm -ivh QConvergeConsoleCLI-AA.BB.CC-DD.Subtype.rpm
```

Linux puts the files in the following directory:

```
/opt/QLogic_Corporation/QConvergeConsoleCLI
```

Red Hat Linux also creates a soft link from the `/usr/local/bin` directory to the executable, `qauc.li`. By default, the `/usr/local/bin` directory is in the execution path; you need not add it.

SUSE Linux/PowerPC® (PPC) does not put the `/usr/local/bin` directory in the execution path by default. You must add it.

## Uninstalling the QConvergeConsole CLI

To remove the QConvergeConsole CLI from your system, follow the instructions for your operating system.

- [Uninstalling the QConvergeConsole CLI in a Windows Environment](#)
- [Uninstalling the QConvergeConsole CLI in a Linux Environment](#)



## Uninstalling the QConvergeConsole CLI in a Windows Environment

The Windows OS offers the following methods of uninstalling the QConvergeConsole CLI:

- [Start Menu Uninstall](#)
- [Control Panel Uninstall](#)
- [Command Line Uninstall](#)

### Start Menu Uninstall

To uninstall from the Windows Start menu:

1. Go to **Start**.
2. Point to **All Programs**, **QLogic Management Suite**, **QConvergeConsole CLI**, and then click **QConvergeConsole CLI**.

### Control Panel Uninstall

To uninstall from the Control Panel:

1. Go to **Start**, and then click **Control Panel**.
2. Click **Add/Remove Programs**.
3. Select **QConvergeConsole CLI**.
4. Click **Change/Remove**.

### Command Line Uninstall

You can uninstall the QConvergeConsole CLI from the command line. Select interactive, passive, or silent (quietly) uninstall:

**To uninstall interactively from the command line:**

Type the following from a command prompt:

```
qcccli.msi
```

**To uninstall passively from the command line:**

Type the following from a command prompt:

```
msiexec /x qcccli.msi
```

**To uninstall quietly from the command line:**

Type the following from a command prompt:

```
msiexec /q /x qcccli.msi
```

---

## Uninstalling the QConvergeConsole CLI in a Linux Environment

To uninstall the QConvergeConsole CLI on a Red Hat/SUSE Linux/PPC operating system, type the following from a command prompt:

```
rpm -e qccccli-AA.BB.CC-DD (be sure to omit the rest of the package name)
```

# 3

## Getting Started

The QConvergeConsole CLI is a unified CLI that manages iSCSI, Ethernet, and Fibre Channel functions on QLogic adapters. The QConvergeConsole CLI noninteractive mode starts, performs the functions defined by the list of parameters provided, and then terminates. Use the noninteractive mode to run the QConvergeConsole CLI from a script file or when you want to perform a single operation.

**NOTE:**

The QConvergeConsole CLI is case sensitive. In addition, file names in some operating systems are case sensitive; in this case, the QConvergeConsole CLI is case sensitive for that particular file.

## Starting the QConvergeConsole CLI

You can start the QConvergeConsole CLI in Windows in the following ways:

- Double click the QCCCLI icon on the desktop.
- Click **Start** and point to **All Programs, QLogic Management Suite**, and then **QConvergeConsole CLI**.
- Open a DOS window, and type the following command:

```
qauc li
```

To start the QConvergeConsole CLI in Linux, type the following command:

```
qauc li
```

## Getting Help

To list all of the available command line parameters, type the following:

```
qauc li -h
```



# 4 NIC Noninteractive Commands

This section describes the noninteractive NIC command format and parameters of the QConvergeConsole CLI.

**NOTE:**

To view help, enter the command **-h**.

## Command Format

Noninteractive mode format includes the following:

```
gauccli -pr nic [optional parameters] <command option> [positional parameters]
```

or

```
gauccli -pr nic -switch [optional cna_port_instance]
```

Where *switch* is one of the following:

- -beacon (Enable/Disable Port Beacon)
- -c
- -cardreset
- -ch
- -d
- -ei
- -g (Display General System Information)
- -h (Help), -?, ?
- -i (List All QLogic CNA Ports Detected)
- -icna (CNA Information)
- -idcbx (Display Port DCBX Information)
- -iset (Display Configured Port Settings)
- -link (Display Physical Link Status)
- -n (Change Port Alias Name)
- -nh (Change CNA Alias Name)
- -pinfo (Port Information)

- -ping (Ping Target)
- -rc
- -sreset (Reset Ethernet Statistics Counters)
- -statport (Display Ethernet Port Statistics)
- -sunreset (Undo Reset Ethernet Statistics Counters)
- -svmtool
- -teamdel (Delete Team)
- -teaminfo (Display Team Information)
- -teamlist (Display Teams List)
- -teamnew (Configure New Team)
- -teamnew\_portspreview
- -testflash
- -testhw
- -testinterrupt
- -testled
- -testlink
- -testregister
- -trans
- -updimages (Update Flash (Boot & Firmware Images))
- -updimages\_viewver
- -v, -ver (Display Program Version Information)
- -vlanadd (Add VLAN to Port or Team)
- -vlanadd\_preview
- -vlandel (Remove VLAN from Port or Team)
- -vlandel\_preview
- -vlaninfo (Display VLAN Information)
- -vlanlist (Display VLAN List)
- -vtcfgview
- -vrestore (Restore VLAN & Teaming Configuration)
- -vtsave (Save VLAN & Teaming Configuration)
- -zvt (Display Teams List; Display VLANs List; Display VLAN Information)
- -zvt (Display Teams List; Display VLANs List; Display VLAN Information)

**NOTE:**

- When you enter a command, the QConvergeConsole CLI loads the adapter, executes the command, and then returns you to the command prompt.
- When optional parameter [cna\_port\_inst] is not present, the command action applies to all adapters detected by the QConvergeConsole CLI.
- Not all switches are functional in this release.
- Ensure that you enter only one command at a time.

## Command Summary

This subsection lists and describes each command line option in alphabetic order, followed by a command description.

### **-beacon** (Enable/Disable Port Beacon)

To toggle on and off the port beacon (LED), issue the `-beacon` command as follows:

```
> -switch -beacon [cna_port_inst] <on:off>
```

For example:

```
> qaucli -pr nic -beacon 2 on
Loading: 1. CNA . . .
Loading: 1. CNA Port index : 1 ...
Loading: 1. CNA Port index : 2 ...
Successfully set Beacon to: Enabled
```

### **-c**

To view the adapter port configuration, issue the `-c` command as follows:

```
> qaucli -pr nic -c [cna_port_inst]
```

For example:

```
> qaucli -pr nic -c 1

=== Displaying CNA Port config info for 1. CNA Port Index ===

Port_Alias                : None
Port_Physical_MAC_Alias    : john
Port_LLA_MAC_Alias         : paul
```

### **-cardreset**

To activate changes made to Flash, issue the `-cardreset` command as follows:

```
> qaucli -pr nic -cardreset [cna_port_inst]
```

### **-ch**

To view the adapter configuration, issue the `-ch` command as follows:

```
> qaucli -pr nic -ch [cna_port_inst]
```

For example:

-d

```
> qaucli -pr nic -ch 1
```

```
=== Displaying CNA config info for 1. CNA Port Index ===
```

```
CNA_Alias                : CNA X54
```

## -d

To generate XML discovery output, issue the -d command as follows:

```
> qaucli -pr nic -d <flash_file> [xml_output_file]
```

where `flash_file` is the Flash file and `xml_output_file` is the XML discovery output. If you omit `xml_output_file`, XML discovery output is written to `stdout.xml`.

## -ei

To get about error and exit code information for noninteractive mode, issue the -ei command as follows:

```
> qaucli -pr nic -ei
```

## -g (Display General System Information)

To view general system information for the host, issue the -g command as follows:

```
> qaucli -pr nic-g [cna_port_inst]
```

For example:

```
> qaucli -pr nic-g 1
```

```
Loading: 1. CNA ...
```

```
Loading: 1. CNA Port index : 1 ...
```

```
Host Name    : STARBLAZER
```

```
Host Version: Service Pack 2 (Build 3790)
```

```
Host Type    : Microsoft Windows Server 2003 R2
```

```
User Type    : Privileged; CNA setup allowed
```



## **-h (Help)**

To view the help file, issue any of the following commands:

```
> qaucli -pr nic -h  
> qaucli -pr nic -?  
> qaucli -pr nic ?
```

The QConvergeConsole CLI help file contains a condensed version of the contents of this chapter.

## **-i (List All QLogic CNA Ports Detected)**

To view a list of all QLogic adapter ports detected, issue the `-i` command as follows:

```
> qaucli -pr nic -i [cna_port_inst]
```

For example:

```
> qaucli -pr nic -i 1
```

```
1. CNA: 1 CNA Port: 1 CNA Model: QLE8142 PortID: 0  
   Mac Phys. Address: 00:c0:dd:12:0f:f4 Loc. Mac: 00:c0:dd:12:0f:f4  
   CNA Serial Number: AFC0916A07152 MPI FW Version: 1.35.00 Adapter Alias:  
None  
   Port Alias: CNA X54  
   IPv4 Address: 192.168.204.201  
   IPv6 Addresses: fe80::2c0:ddff:fe12:ff4
```

## **-icna (CNA Information)**

To view adapter information, issue the `-icna` command as follows:

```
> qaucli -pr nic -icna [cna_port_inst]
```

For example:

```
> qaucli -pr nic -icna 2
```

```
=== CNA Information for 2. CNA Port Index ===  
Hostname                : WIN-4ZKSPMU5N5L  
Adapter Model           : QLE8142  
Chip Model              : 8001  
Chip Version            : A1
```

```
Adapter Alias           : None
Serial Number           : RFC0916K61116
MAC Address              : 00:c0:dd:10:27:60
MAC Address              : 00:c0:dd:10:27:62
Driver Information       : Ndis 6.x 10GbE driver (X86)
Driver Name              : qlge.sys
Driver Version           : 1.0.1.0
MPI Firmware Version     : 1.35.06
PXE Boot Version         : NA
VLAN & Team Driver Name  :
VLAN & Team Driver Version :
FCoE Driver Version      : 9.1.8.17
FCoE Firmware Version    : 5.01.03
FCoE SDMAPI Version      : 01.28.00.75
```

## **-idcbx (Display Port DCBX Information)**

To display port data center bridging exchange protocol (DCBX) and enhanced transmission selection (ETS) information, issue the `-idcbx` command as follows:

```
> qauccli -pr nic -idcbx [cna_port_inst]
```

For example:

```
> qauccli -pr nic -idcbx 1
```

```
-----
DCBX
```

```
-----
DCBX Enable           : Enabled
DCBX Negotiation      : Successful
Port Pause Type       : Per Priority Pause
```

```
-----
ETS
```

```
-----
Priority Tx Mode       : Bandwidth
SAN Unused BW to LAN  : Disabled
LAN Unused BW to SAN  : Enabled
```

```
Class 1 Priority Group ID : 1
```

```
Class 1 Bandwidth Percent : 1
Class 1 CoS Priorities    : 3

Class 2 Priority Group ID : 2
Class 2 Bandwidth Percent : 99
Class 2 CoS Priorities    : 0 1 2 4 5 6 7
```

## **-intloopback (Configure Test Parameters)**

To configure loopback test parameters before you run the test, issue the `intloopback` command as follows:

```
> qauccli -pr nic intloopback <cna_port_inst> <tests_num>
<on_error>
```

Where `tests_num` specifies the number of tests to run (a value between 1 and 65,535) and `on_error` specifies the action to take if an error is encountered in the test (0 = ignore the error; 1 = abort the test).

## **-iset (Display Configured Port Settings)**

To view configured port settings, issue the `-iset` command as follows:

```
> qauccli -pr nic -iset [cna_port_inst]
```

For example:

```
> qauccli -pr nic -iset 1
```

```
=== Configured Port Settings for 1. CNA Port Index ===
Header_Data_Split_Enable          : off
IPv4_Large_Send_Offload_v1_Enable : on
IPv4_Large_Send_Offload_v2_Enable : on
IPv6_Large_Send_Offload_v2_Enable : on
IPv4_TCP_Checksum_Offload_Enable  : RxTx
IPv6_TCP_Checksum_Offload_Enable  : RxTx
IPv4_UDP_Checksum_Offload_Enable  : RxTx
IPv6_UDP_Checksum_Offload_Enable  : RxTx
Jumbo_Frames_MTU_9000_Enable_Rx   : off
Jumbo_Frames_MTU_9000_Enable_Tx   : off
LOCAL_Administered_Address_MAC    : 00:c0:dd:12:0f:f4
MAC_Address_Alias                  : paul
Receive_Side_Scaling_Enable       : on
Receive_Ring_Size                  : 0
```

```
Transmit_Ring_Size          : 0
PromiscuousModeEnabled      : off
```

## **-link** **(Display Physical Link Status)**

To view the status of the physical link, including the port speed, issue the `-link` command as follows:

```
> qaucli -pr nic -link [cna_port_inst]
```

For example:

```
> qaucli -pr nic -link 1
```

```
=== Physical Link Status for 1. CNA Port Index ===
Link Status          : Up
Port Speed           : 10 Gbps
```

## **-n** **(Change Port Alias Name)**

To change port alias names, issue the `-n` command as follows:

```
> qaucli -pr nic -n [cna_port_inst] <config_name|config_alias>
<value> [<config_name|config_alias> <value>]
```

For this release of the QConvergeConsole CLI, you can set the following properties:

- Port\_Alias
- Port\_Physical\_MAC\_Alias
- Port\_LLA\_MAC\_Alias

For example:

```
> qaucli -pr nic -n 1 Port_Alias jwm614
```

```
Port_Alias           : jwm614
```

## **-nh** **(Change CNA Alias Name)**

To configure the adapter, issue the `-nh` command as follows:

```
> qaucli -pr nic -nh [cna_port_inst] <config_name|config_alias>
<value> [<config_name|config_alias> <value>]
```

For this release, you can set the only the `CNA_Alias` property.

For example:

```
> qauccli -pr nic -nh 1 CNA_Alias george
```

```
CNA_Alias : george
```

## **-pinfo (Port Information)**

To view port information, issue the `-pinfo` command as follows:

```
> qauccli -pr nic -pinfo [cna_port_inst]
```

For example:

```
> qauccli -pr nic -pinfo
```

```
=== Port Information for 1. CNA Port Index ===
```

```

Hostname : STARBLAZER
Adapter Model : QLE8142
Port Alias : None
Physical MAC Address : 00:c0:dd:12:0f:f4
Physical MAC Address Alias : paul
Configured (LLA) MAC Address: 00:c0:dd:12:0f:f4
LAA MAC Address Alias : paul
IPv4 Address : 192.168.204.201
IPv4 Subnet Mask : 255.255.255.0
IPv4 Default Gateway : 255.255.255.255
IPv4 DHCP Enabled : No
DNS Servers : fec0:0:0:ffff::1
              : fec0:0:0:ffff::2
              : fec0:0:0:ffff::3
IPv6 Addresses : fe80::2c0:ddff:fe12:ff4
                : fec0:0:0:ffff::1
                : fec0:0:0:ffff::2
IPv6 Default Router : fe80::2c0:ddff:fe12:ff4
Link Status : Up
MTU : 1514
Interface Scope ID : 91
Interface Speed : 10 Gbps
Interface Description : QLogic 10Gb PCI Ethernet Adapter -
Network Load Balancing Filter Device
```

```

=== Port Information for 2. CNA Port Index ===
. . .

```

## **-ping (Ping Target)**

To run a ping diagnostic, issue the `-ping` command as follows:

```

> qauccli -pr nic -ping <cna_port_inst> <hostname_or_IPv4> [<count>
<packet_size> <timeout_ms> <TTL>]

```

Where the default parameter values are as follows:

```

count = 5
packet size = 525
timeout_ms = 1000
TTL = 30

```

For example:

```

> qauccli -pr nic -ping 1 starblazer
Loading: 1. CNA ...
Loading: 1. CNA Port index : 1 ...
Loading: 1. CNA Port index : 2 ...
Got IP: 192.168.104.142 for starblazer
-----
ping IPv4
-----
Got IP: 192.168.104.142 for starblazer
Pinging 192.168.104.142 ...
Ping Response Received, roundtrip=0 ms, hops=0
Ping Response Received, roundtrip=0 ms, hops=0
Ping Response Received, roundtrip=0 ms, hops=0
Ping Response Received, roundtrip=0 ms, hops=0
Ping Response Received, roundtrip=16 ms, hops=0
Summary: ping failed 0, ping success 5 ping timeout 0

```

## **-rc**

To view a list of error codes and their descriptions, issue the `-rc` command as follows:

```

> qauccli -pr nic -rc

```

## **-sreset** **(Reset Ethernet Statistics Counters)**

To reset the Ethernet statistics counters, issue the `-sreset` command as follows:

```
> qauccli -pr nic -sreset [cna_port_inst]
```

For example:

```
> qauccli -pr nic -sreset 1
```

Port statistics reset for 1. CNA Port Index

## **-statport** **(Display Ethernet Port Statistics)**

To view the Ethernet port statistics, issue the `-statport` command as follows:

```
> qauccli -pr nic -statport [cna_port_inst]
```

For example:

```
> qauccli -pr nic -statport 1
```

Port statistics for 1. CNA Port Index

|                        |           |
|------------------------|-----------|
| txPkts                 | : 12989   |
| txOctets               | : 2302620 |
| txMulticastPkts        | : 8444    |
| txBroadcastPkts        | : 4195    |
| txUnicastPkts          | : 350     |
| txControlPkts          | : 0       |
| txPausePkts            | : 0       |
| txPkts64Octets         | : 1013    |
| txPkts65to127Octets    | : 8746    |
| txPkts128to255Octets   | : 422     |
| txPkts256to511Octets   | : 2458    |
| txPkts512to1023Octets  | : 0       |
| txPkts1024to1518Octets | : 350     |
| txPkts1519toMaxOctets  | : 0       |
| txUndersizePkts        | : 0       |
| txOversizePkts         | : 0       |
| rxOctets               | : 3051428 |
| rxPkts                 | : 18724   |
| rxBroadcastPkts        | : 5728    |

```
rxMulticastPkts      : 12646
rxUnicastPkts        : 350
rxUndersizePkts      : 0
rxOversizePkts       : 0
rxJabberPkts         : 0
rxUndersizeFCSErrorPkts : 0
rxPkts640ctets       : 417
rxPkts65to1270ctets  : 9351
rxPkts128to2550ctets : 8252
rxPkts256to5110ctets : 354
rxPkts512to10230ctets : 0
rxPkts1024to15180ctets : 350
rxPkts1519toMax0ctets : 0
rxControlPkts        : 0
rxPausePkts          : 0
txCBFCPauseFrames0   : 0
txCBFCPauseFrames1   : 0
txCBFCPauseFrames2   : 0
txCBFCPauseFrames3   : 0
txCBFCPauseFrames4   : 0
txCBFCPauseFrames5   : 0
txCBFCPauseFrames6   : 0
txCBFCPauseFrames7   : 0
txFCoEPkts           : 306
txMgmtPkts           : 7921
rxCBFCPauseFrames0   : 0
rxCBFCPauseFrames1   : 0
rxCBFCPauseFrames2   : 0
rxCBFCPauseFrames3   : 0
rxCBFCPauseFrames4   : 0
rxCBFCPauseFrames5   : 0
rxCBFCPauseFrames6   : 0
rxCBFCPauseFrames7   : 0
rxFCoEPkts           : 0
rxMgmtPkts           : 7912
rxPktsPriority0       : 0
rxPktsPriority1       : 0
rxPktsPriority2       : 0
rxPktsPriority3       : 306
```



```

rxPktsPriority4      : 0
rxPktsPriority5      : 0
rxPktsPriority6      : 0
rxPktsPriority7      : 0
txPktsPriority0      : 0
txPktsPriority1      : 0
txPktsPriority2      : 0
txPktsPriority3      : 306
txPktsPriority4      : 0
txPktsPriority5      : 0
txPktsPriority6      : 0
txPktsPriority7      : 4762
rxPktsDiscardPriority0 : 0
rxPktsDiscardPriority1 : 0
rxPktsDiscardPriority2 : 0
rxPktsDiscardPriority3 : 0
rxPktsDiscardPriority4 : 0
rxPktsDiscardPriority5 : 0
rxPktsDiscardPriority6 : 0
rxPktsDiscardPriority7 : 0

```

## **-sunreset (Undo Reset Ethernet Statistics Counters)**

To undo the resetting of Ethernet statistics counters, issue the `-sunreset` command as follows:

```
> qauccli -pr nic -sunreset [cna_port_inst]
```

For example:

```
> qauccli -pr nic -sunreset 1
```

Port statistics undo reset for 1. CNA Port Index

## **-svmtool**

The SVM tool is a general purpose utility that displays and configures settings for QLogic Converged Network Adapters.

To update the SVM firmware, issue the `-svmtool` command as follows:

```
> qauccli -pr nic -svmtool [options]
```

where [options] are:

mode=[version|ver|inventory|update]—mode of execution  
 fwup=fw\_file\_name—file containing firmware  
 output=[stdout|stderr|output\_file\_name]—output in XML  
 format; can be repeated  
 mac=[11:22:33:44:55:66|11-22-33-44-55-66]—limits update to  
 device with physical MAC address. This option applies only when  
 mode=update.  
 fwdump=fwdump\_file\_name—not applicable  
 comment=[on|true]—inserts comments into XML output on MAC  
 address of the device

For example, to update firmware from a file, produce XML output to stdout and to a file, and add comments to XML on MAC of the device, issue the `-svmtool` command as follows:

```
> qauccli -pr nic -svmtool mode=update
fwup=c:/fw/fw1.bin output=stdout
output=c:/xmloutput/1.xml comment=on
```

## **-teamdel** (Delete Team)

To delete a team, issue the `-teamdel` command as follows:

```
> qauccli -pr nic -teamdel <team_inst|ALL>
```

For example:

```
> qauccli -pr nic -teamdel 3
```

About to delete team: 1 (QLogic VT-IM Miniport Driver #2). Please wait ...

Successfully deleted team: 1 ( )

## **-teaminfo** (Display Team Information)

To view information about a team, issue the `-teaminfo` command as follows:

```
> qauccli -pr nic -teaminfo <team_inst|ALL>
```

For example:

```
> qauccli -pr nic -teaminfo ALL
```

```
***** Team: QLogic VT-IM Miniport Driver #2 *****
```

Team Description: QLogic VT-IM Miniport Driver #2

```
Team Type      : Fail Over
Driver Name    : qlvtid.sys
Driver Version : 1.0.0.2
Driver Date    : 05/26/2009
VLAN Enabled   : Disabled
VLAN ID        : None
MAC Address    : 00:c0:dd:0a:b1:a9
MTU            : 1500
IPv4 Address   : 169.254.27.115
Subnet Mask    : 255.255.0.0
IPv6 Address   : fe80::2c0:ddff:fe0a:b1a9
Link Status    : Unknown
```

## **-teamlist** (Display Teams List)

To view a list of all previously configured teams (if any), issue the `-teamlist` command as follows:

```
> qaucli -pr nic -teamlist
```

For example:

```
> qaucli -pr nic -teamlist
```

Using config file: `.\netscli.cfg`

Loading: 1. CNA ...

Loading: 1. CNA Port index : 1 ...

Loading: 1. CNA Port index : 2 ...

Team: 1 Team Description: QLogic VT-IM Miniport Driver #2 VLAN ID: None  
Team Type: Fail Over

Team Members:

CNA: 1 CNA Port: 1 MAC: 00:c0:dd:0a:b1:a8 Description: QLogic 10Gb PCI  
Ethernet Adapter

CNA: 1 CNA Port: 2 MAC: 00:c0:dd:0a:b1:a9 Description: QLogic 10Gb PCI  
Ethernet Adapter #2

## **-teamnew** (Configure New Team)

To configure a new team, issue the `-teamnew` command as follows:

```
> qaucli -pr nic -teamnew <team_type> <port_insts|ALL>
```

Where `team_type` specifies a numeric value and `port_insts` specifies a list of comma-separated port indices (for example 1, 2).

`team_type` can have the following values:

- 1 = Fail-safe team
- 2 = Switch independent load balancing
- 3 = 802.3ad static team
- 4 = 802.3ad dynamic team—active link aggregation control protocol (LACP)
- 5 = 802.3ad dynamic team—passive LACP

For example:

```
> qauccli -pr nic -teamnew 1 ALL
```

User selected CNA Port Indices: ALL

Attempting to create new team:

Team Description:

Team Type: Fail Over

Selected ports : 1, 2

CNA: 1 CNA Port: 1 MAC: 00:c0:dd:0a:b1:a8 Description: QLogic 10Gb PCI Ethernet Adapter - Network Load Balancing Filter Device

QLogic 10Gb PCI Ethernet Adapter - Network Load Balancing Filter Device

CNA: 1 CNA Port: 2 MAC: 00:c0:dd:0a:b1:a9 Description: QLogic 10Gb PCI Ethernet Adapter #2 - Network Load Balancing Filter Device

QLogic 10Gb PCI Ethernet Adapter #2 - Network Load Balancing Filter Device

About to create the team. Please wait ...

Successfully created team with interface description: QLogic VT-IM Miniport Driver #2

## **-teamnew\_portspreview**

To preview ports before you configure a new team, issue the `-teamnew_portspreview` command as follows:

```
> qauccli -pr nic -teamnew <team_type> <port_insts|ALL>
```

For example:

```
> qaucli -pr nic -teamnew 1 ALL
```

```
1. CNA: 1 CNA Port: 1 CNA Model: QLE8142 PortID: 0
   Mac Phys. Address: 00:c0:dd:12:0f:f4 Loc. Mac: 00:c0:dd:12:0f:f4
   CNA Serial Number: AFC0916A07152 MPI FW Version: 1.35.02 Adapter Alias: None
   Port Alias: None
   IPv4 Address: 192.168.204.201
   IPv6 Addresses: fe80::2c0:ddff:fe12:ff4
2. CNA: 1 CNA Port: 2 CNA Model: QLE8142 PortID: 1
   Mac Phys. Address: 00:c0:dd:12:0f:f6 Loc. Mac: 00:c0:dd:12:0f:f6
   CNA Serial Number: AFC0916A07152 MPI FW Version: 1.35.02 Adapter Alias: None
   Port Alias: None
   IPv4 Address: 192.168.205.200
   IPv6 Addresses: fe80::2c0:ddff:fe12:ff6
```

## -testflash

To test the Flash memory, issue the -testflash command as follows:

```
> qaucli -pr nic -testflash [cna_port_inst]
```

## -testhw

To test the hardware, issue the -testhw command as follows:

```
> qaucli -pr nic -testhw [cna_port_inst]
```

## -testinterrupt

To test the interrupt, issue the -testinterrupt command as follows:

```
> qaucli -pr nic -testinterrupt [cna_port_inst]
```

## -testled

To test the LED, issue the -testled command as follows:

```
> qaucli -pr nic -testled [cna_port_inst]
```

## -testlink

To test the link, issue the -testlink command as follows:

```
> qaucli -pr nic -testlink [cna_port_inst]
```

## -testregister

To test the register, issue the -testregister command as follows:

```
> qaucli -pr nic -testregister [cna_port_inst]
```

## **-trans**

Display transceiver FDMI information.

## **-updimages (Update Flash (Boot & Firmware Images))**

To update the Flash (boot and firmware) images, issue the `-updimages` command as follows:

```
> qauccli -pr nic -updimages [cna_port_inst] <image_file>
```

For example:

```
> qauccli -pr nic -updimages 1 "c:\my_dir\valid-image-file"
netscli -updimages c:\temp\temp1
Loading: 1. CNA ...
Loading: 1. CNA Port index : 1 ...
Loading: 1. CNA Port index : 2 ...
=== Updating images for 1. CNA Port Index ===
Loading, please wait ...
Successfully updated Flash (Boot & Firmware images) for 1. Port
Index from file: c:\temp\temp1
=== Updating images for 2. CNA Port Index ===
Loading, please wait ...
Successfully updated Flash (Boot & Firmware images) for 2. Port
Index from file: c:\temp\temp1
```

## **-updimages\_viewver**

To view the Flash version, issue the `-updimages_viewver` command as follows:

```
> qauccli -pr nic -updimages_viewver image_file
```

## **-v, -ver (Display Program Version Information)**

To view the version number of the QConvergeConsole CLI, issue the `-v` or `-ver` command as follows:

```
> qauccli -pr nic -v
> qauccli -pr nic -ver
```

For example:

```
> qauccli -pr nic -ver
```

QConvergeConsole CLI Utility

Program Version : 1.0.00.xx  
netSDMAPI : 00.00.58  
Copyright 1999-2009 QLogic Corp.

## **-vlanadd (Add VLAN to Port or Team)**

To add a virtual LAN (VLAN) to a port or team, issue the `-vlanadd` command as follows:

```
> gauccli -pr nic -vlanadd <port_insts> <vlan_id>
```

Where `port_insts` specifies the port's indices comma separated (for example, 1,2) and `vlan_id` specifies a numeric value 1–4095.

For example:

```
> gauccli -pr nic -vlanadd 2 120
```

Using config file: `.\netscli.cfg`

Loading: 1. CNA ...

Loading: 1. CNA Port index : 1 ...

Loading: 1. CNA Port index : 2 ...

VLAN List:

ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC:  
00:c0:dd:0a:b1:a8 Description: QLogic 10Gb PCI Ethernet Adapter -  
Network Load Balancing Filter Device

ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC:  
00:c0:dd:0a:b1:a9 Description: QLogic 10Gb PCI Ethernet Adapter #2  
- Network Load Balancing Filter Device

User entered ListIndices: 2

User entered VLAN ID: 120

About to set VLAN ID: 120 for ListIndex: 2

Successfully set.

## **-vlanadd\_preview**

To preview the VLAN, port, and team information before you add a VLAN to a port or team, issue the `-vlanadd_preview` command as follows:

```
> gauccli -pr nic -vlanadd_preview
```

For example:

```
> gauccli -pr nic -vlanadd_preview
```

VLAN List:

```
ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC:
00:c0:dd:0a:b1:a8 Description: QLogic 10Gb PCI Ethernet Adapter -
Network Load Balancing Filter Device
ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC:
00:c0:dd:0a:b1:a9 Description: QLogic 10Gb PCI Ethernet Adapter #2
- Network Load Balancing Filter Device
```

## **-vlandel (Remove VLAN from Port or Team)**

To remove all VLANS from a port or team, issue the `-vlandel` command as follows:

```
> qaucli -pr nic -vlandel <port_insts|ALL> <vlan_id|ALL>
```

Where `port_insts` specifies the port's indices comma separated (for example, 1,2) and `vlan_id` specifies a numeric value 1–4095.

For example:

```
> qaucli -pr nic -vlandel ALL ALL
```

VLAN List:

```
ListIndex: 1 CNA: 1 CNA Port: 2 VLAN ID: 120 MAC:
00:c0:dd:0a:b1:a9 Description: QLogic 10Gb PCI Ethernet Adapter #2
```

User entered ListIndices: ALL

User entered VLAN ID: ALL

About to remove VLAN ID(s) for ListIndex: 1. (QLogic 10Gb PCI Ethernet Adapter #2) Please wait ...

Successfully removed VLAN ID.

## **-vlandel\_preview**

To preview the virtual LAN, port, and team information before you delete a VLAN from a port or team, issue the `-vlandel_preview` command as follows:

```
> qaucli -pr nic -vlandel_preview
```

For example:

```
> qaucli -pr nic -vlandel_preview
```

VLAN List:

```
ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: 453 MAC:
00:c0:dd:12:0f:f4 Description: QLogic 10Gb PCI Ethernet Adapter
```



## **-vlaninfo** (Display VLAN Information)

To view information about the VLAN, issue the `-vlaninfo` command as follows:

```
> qaucli -pr nic -vlaninfo <vlan_inst|ALL>
```

For example:

```
> qaucli -pr nic -vlaninfo 1
```

VLAN(s) Information:

Updating VLANs info ...

Done ...

No VLANs to display.

## **-vlanlist** (Display VLAN List)

To view the adapters, ports, teams, VLAN IDs, and descriptions, issue the `-vlanlist` command as follows:

```
> qaucli -pr nic -vlanlist
```

For example:

```
> qaucli -pr nic -vlanlist
```

VLAN List:

CNA: 1 CNA Port: 2 VLAN ID: 120 MAC: 00:c0:dd:0a:b1:a9

Description: QLogic VT-IM Miniport Driver #2

## **-vtcfgview**

To view the contents of the previously saved VLAN and teaming configuration settings file (see [-vtsave \(Save VLAN & Teaming Configuration\)](#)), issue the `-vtcfgview` command as follows:

```
> qaucli -pr nic -vtcfgview [state_cfg_file]
```

### **NOTE:**

If you do not specify a file name, the QConvergeConsole CLI reads the default file (`vtstate.cfg`) in the installation directory

For example:

```
>gauccli -pr nic -vtcfgview c:/system_1.cfg
```

```
Using config file: C:\Program Files (x86)\QLogic
Corporation\SANsurferNICCLI\netscli.cfg
```

```
Accessing file: c:/system_1.cfg
```

```
Number of Teams: 1
```

```
-----
```

```
type=Fail Over MAC: 00:c0:dd:0a:b4:61 00:c0:dd:0a:b4:60 (VLAN IDs: 555)
```

```
Number of Ports: 3
```

```
-----
```

```
MAC :00:c0:dd:0a:b4:34 (VLAN IDs: 35)
```

```
MAC :00:c0:dd:0a:b4:35 (VLAN IDs: 777)
```

```
MAC :None (VLAN IDs:)
```

```
Number of VLANs: 3
```

```
-----
```

```
.....
VLAN ID          : 35
.....
IPv4Address       : 169.254.132.7
IPv4SubnetMask    : 0.0.0.0
IPv4GatewayAddr   : 0.0.0.0
DHCPServerCount   : 0
IPv6GatewayAddr   :
IPv6AddressCount  : 1
pDHCPServers[    0] : fe80::e9ba:1d3e:8584:8407
DNSServerCount    : 3
pDNSServers[     0] : fec0:0:0:ffff::1
pDNSServers[     1] : fec0:0:0:ffff::2
pDNSServers[     2] : fec0:0:0:ffff::3
.....
```

```
VLAN ID          : 777
.....
IPv4Address       : 169.254.202.39
IPv4SubnetMask    : 0.0.0.0
IPv4GatewayAddr   : 0.0.0.0
DHCPServerCount   : 0
IPv6GatewayAddr   :
```

```
IPv6AddressCount      : 1
pDHCPservers[ 0]      : fe80::d079:f594:8e03:ca27
DNSServerCount        : 3
pDNSServers[ 0]       : fec0:0:0:ffff::1
pDNSServers[ 1]       : fec0:0:0:ffff::2
pDNSServers[ 2]       : fec0:0:0:ffff::3
.....
VLAN ID               : 555
.....
IPv4Address           :
IPv4SubnetMask         :
IPv4GatewayAddr       :
DHCPserverCount        : 0
IPv6GatewayAddr       :
IPv6AddressCount       : 0
DNSServerCount         : 0
```

## **-vtrestore (Restore VLAN & Teaming Configuration)**

To reinstate previously saved VLAN and team settings from a file, issue the -vtrestore command as follows:

```
> qauccli -pr nic -vtrestore [file_name]
```

The default file configuration file name for VLAN and teaming is vtstate-user.cfg, which is saved in the QConvergeConsole CLI installation directory. You may have saved your configuration with a different file name or location.

### **NOTE:**

The -vtrestore command can restore a configuration file produced only by the -vtsave command. Similarly, a configuration that is saved using the QLogic Windows Teaming Property pages can be restored only with the Windows Teaming Property Pages.

For example:

```
>qauccli -pr nic -vtrestore c:/system_1.cfg
Using config file: C:\Program Files (x86)\QLogic
Corporation\SANsurferNICCLI\netscli.cfg
Loading CNA Data ...
Loading: 1. CNA ...
Loading: 1. CNA Port index : 1 ...
Loading: 1. CNA Port index : 2 ...
Loading: 2. CNA ...
Loading: 2. CNA Port index : 1 ...
Loading: 2. CNA Port index : 2 ...
Refreshing interfaces ... Please wait ...
Attempting to restore TEAMS and VLANs if configured prior to vtdriver update.
*** Teams:
*** Ports:
00:c0:dd:0a:b4:34
VLAN IDs:
55

00:c0:dd:0a:b4:35
VLAN IDs:
55

Refreshing interfaces ... Please wait ...
Updating IP properties for all ports ... Please wait ...
ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:60
Description: QLogic 10Gb PCI Ethernet Adapter #3
ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:61
Description: QLogic 10Gb PCI Ethernet Adapter #4
ListIndex: 3 CNA: 2 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:34
Description: QLogic 10Gb PCI Ethernet Adapter
ListIndex: 4 CNA: 2 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:35
Description: QLogic 10Gb PCI Ethernet Adapter #2

VLAN List:
ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:60
Description: QLogic 10Gb PCI Ethernet Adapter #3
ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:61
Description: QLogic 10Gb PCI Ethernet Adapter #4
ListIndex: 3 CNA: 2 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:34
Description: QLogic 10Gb PCI Ethernet Adapter
ListIndex: 4 CNA: 2 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:35
```

Description: QLogic 10Gb PCI Ethernet Adapter #2

Selected ListIndices: 3

Selected VLAN ID: 55

About to set VLAN ID: 55 for ListIndex: 3

Successfully set VLAN ID.

Refreshing interfaces ... Please wait ...

Updating IP properties for all ports ... Please wait ...

ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:60  
Description: QLogic 10Gb PCI Ethernet Adapter #3

ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:61  
Description: QLogic 10Gb PCI Ethernet Adapter #4

ListIndex: 3 CNA: 2 CNA Port: 1 VLAN ID: 55 MAC: 00:c0:dd:0a:b4:34  
Description: QLogic 10Gb PCI Ethernet Adapter

ListIndex: 4 CNA: 2 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:35  
Description: QLogic 10Gb PCI Ethernet Adapter #2

VLAN List:

ListIndex: 1 CNA: 1 CNA Port: 1 VLAN ID: None MAC: 00:c0:dd:0a:b4:60  
Description: QLogic 10Gb PCI Ethernet Adapter #3

ListIndex: 2 CNA: 1 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:61  
Description: QLogic 10Gb PCI Ethernet Adapter #4

ListIndex: 3 CNA: 2 CNA Port: 1 VLAN ID: 55 MAC: 00:c0:dd:0a:b4:34  
Description: QLogic 10Gb PCI Ethernet Adapter

ListIndex: 4 CNA: 2 CNA Port: 2 VLAN ID: None MAC: 00:c0:dd:0a:b4:35  
Description: QLogic 10Gb PCI Ethernet Adapter #2

Selected ListIndices: 4

Selected VLAN ID: 55

About to set VLAN ID: 55 for ListIndex: 4

Successfully set VLAN ID.

Refreshing interfaces ... Please wait ...

Updating IP properties for all ports ... Please wait ...

Successfully restored TEAMS/VLANs

## **-vtsave** **(Save VLAN & Teaming Configuration)**

To save the currently configured VLAN and team settings to a file, issue the  
-vtsave command as follows:

> **gauccli -pr nic -vtsave [file\_name]**

**NOTE:**

- If you do not specify a configuration file name, the VLAN and teaming settings are saved in the default file, vtstate-user.cfg, which is saved in the QConvergeConsole CLI installation directory. Saved settings can be later reinstated using the -vtrestore option.
- A configuration that is saved with the -vtsave command can be restored only with the -vtrestore command. Similarly, a configuration that is saved using the QLogic Windows Teaming Property pages can be restored only with the Windows Teaming Property Pages.

```
>qauccli -pr nic -vtsave c:/system_1.cfg
```

```
Using config file: C:\Program Files (x86)\QLogic  
Corporation\SANsurferNICCLI\netscli.cfg
```

```
Loading CNA Data ...
```

```
Loading: 1. CNA ...
```

```
Loading: 1. CNA Port index : 1 ...
```

```
Loading: 1. CNA Port index : 2 ...
```

```
Loading: 2. CNA ...
```

```
Loading: 2. CNA Port index : 1 ...
```

```
Loading: 2. CNA Port index : 2 ...
```

```
Refreshing interfaces ... Please wait ...
```

```
Attempting to save TEAMS and VLANs Configuration.
```

```
Team: 1 Team Description: QLogic VT-IM Miniport Driver VLAN ID: 555 Team Type:  
Fail Over
```

```
Team Members:
```

```
CNA: 1 CNA Port: 2 MAC: 00:c0:dd:0a:b4:61 Description: QLogic 10Gb PCI  
Ethernet Adapter #4
```

```
CNA: 1 CNA Port: 1 MAC: 00:c0:dd:0a:b4:60 Description: QLogic 10Gb PCI  
Ethernet Adapter #3
```

```
VLAN List:
```

```
ListIndex: 1 Team: 1 Team Description: QLogic VT-IM Miniport Driver VLAN ID:  
555 MAC: 00:c0:dd:0a:b4:60 Description: QLogic VT-IM Miniport Driver
```

```
VLAN List:
```

```
Team: 1 Team Description: QLogic VT-IM Miniport Driver VLAN ID: 555 MAC:  
00:c0:dd:0a:b4:60 Description: QLogic VT-IM Miniport Driver #2
```

```
Successfully saved TEAMS/VLANs
```

## **-zvt (Display Teams List; Display VLANs List; Display VLAN Information)**

To view a display that combines all VLAN and teaming information, issue the -zvt command as follows:

```
> qaucli -pr nic -zvt
```

The following example shows the output on a system with one team of two ports, plus two additional ports:

```
> qaucli -pr nic -zvt
```

```
*****
*****
*** Displaying VLAN & Teaming Information ***
*****
*****

*** Display Of Teams List ***

Team: 1 Team Description: QLogic VT-IM Miniport Driver VLAN ID: 555 Team Type:
Fail Over
Team Members:
CNA: 1 CNA Port: 2 MAC: 00:c0:dd:0a:b4:61 Description: QLogic 10Gb PCI
Ethernet Adapter #4
CNA: 1 CNA Port: 1 MAC: 00:c0:dd:0a:b4:60 Description: QLogic 10Gb PCI
Ethernet Adapter #3

*** Display Of VLANs List ***
VLAN List:

Team: 1 Team Description: QLogic VT-IM Miniport Driver VLAN ID: 555 MAC:
00:c0:dd:0a:b4:60 Description: QLogic VT-IM Miniport Driver #2
*** Display Of VLAN Information ***

VLAN(s) Information:
Updating VLANs info ...
Done ...
Available VLAN ID(s):
555
```

\*\*\*\*\* VLAN: 555 (QLogic VT-IM Miniport Driver #2) \*\*\*\*\*

Driver Name : qlvtid.sys  
Driver Version : 1.0.0.14  
Driver Date : 10/13/2009  
VLAN Enabled : Enabled  
VLAN ID : 555  
MAC Address : 00:c0:dd:0a:b4:60  
MTU : 1514  
IPv4 Address :  
Subnet Mask :  
IPv6 Address :  
Link Status : Down



# 5 Fibre Channel Noninteractive Commands

This section describes the noninteractive Fibre Channel command format and parameters of the QConvergeConsole CLI. For a quick reference to informative commands, see [Display System Information \(Command Line Options -g, -z, and -tp\)](#).

**NOTE:**

To view help, enter the command **-h**.

## Display System Information (Command Line Options -g, -z, and -tp)

When you enter one of these options, general information appears in various formats. For example:

- `# qauccli -pr fc -g` shows the host information (see [“Host Information \(Command Line Option -g\)”](#) on page 5-1).
- `# qauccli -pr fc -z` shows the host configuration (see [“Host Configuration \(Command Line Option -z\)”](#) on page 5-2).

## Host Information (Command Line Option -g)

The command format to show host information is:

```
# qauccli -pr fc -g
```

When you enter this option, the following information about the local machine appears:

- Host name
- OS type
- OS version (patches where applicable)
- SAN target management (SDM) API version

- List of adapters: adapter model, port number, WWPN, serial number (SN), adapter number (adapter 0–n), and status (online/offline)
- Total number of QLogic Fibre Channel adapters detected

**NOTE:**

The failover and SAN device management APIs are QLogic-specific libraries required for the QConvergeConsole CLI. The versions of these libraries are useful for debugging purposes.

## Host Configuration (Command Line Option -z)

The command line option -z provides a summary for the selected adapter in a single command. The command format is:

```
# qaucli -pr fc -z
```

The QConvergeConsole CLI shows the information in [“Host Information \(Command Line Option -g\)” on page 5-1](#), as well as the following additional information:

- Adapter general information
- Adapter vital product data (VPD)
- Adapter parameter settings
- Driver settings information:
  - ☐ Group: persistent
  - ☐ Group: binding
- Device/LUN Information
- Selective LUNs Information
  - ☐ Boot device settings

To show the information for a single adapter, type the following command:

```
# qaucli -pr fc -z (<hba instance> | <hba wwpn>)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

To show the information for all adapters, type:

```
# qaucli -pr fc -z all
```

## Host Configuration

Use these options to show the information for one or all of the adapter ports in the system. The -z option shows the combination of the commands listed in [Table 5-1](#). The commands are listed in alphabetical order.

**Table 5-1. Options Shown by -z**

| CLI | Description                         | See User Guide Section                                                                                     |
|-----|-------------------------------------|------------------------------------------------------------------------------------------------------------|
| -c  | Show adapter settings               | <a href="#">“-c (Show Parameter Settings)” on page 5-7</a>                                                 |
| -e  | Configure boot device               | <a href="#">“-e (Boot Device View   Select   Disable)” on page 5-9</a>                                     |
| -fg | View driver settings                | <a href="#">“-fg (Show Driver Settings)” on page 5-12</a>                                                  |
| -g  | Show system information             | <a href="#">“-gs (Show HBA Statistics)” on page 5-13</a>                                                   |
| -i  | Show adapter information            | <a href="#">“-i (Display HBA General Information or VPD Information for One or All HBAs)” on page 5-16</a> |
| -l  | Show LUN list                       | <a href="#">“-l (Display LUN Info)” on page 5-21</a>                                                       |
| -m  | Configure selective LUNs            | <a href="#">“-m (View, Enable, or Disable LUNs)” on page 5-22</a>                                          |
| -p  | Configure target persistent binding | <a href="#">“-p (Display Persistent Target Binding Info)” on page 5-26</a>                                 |
| -t  | Show target list                    | <a href="#">“-t (Display Target Information)” on page 5-32</a>                                             |

## Command Format

The command format for noninteractive mode is:

```
# qaucli -pr fc -(Cmd) (<hba instance> | <hba wwpn>)
(view | ?)
```

Where:

*hba instance* = Adapter port instance (use -g command to find)  
*hba wwpn* = Adapter world wide port name

You can combine the commands -o, -s and -x with other options. However, they *must* be at the beginning or at the end of the command line. Use these commands as follows:

- -o = Output the results to a file (see [“-o \(Output to a File\)” on page 5-26](#)). For example:  
# **qauccli -pr fc -l -o**
- -s = Silent mode (see [“-s \(Silent Mode\)” on page 5-31](#)). For example:  
# **qauccli -pr fc -i all -s -o output.txt**
- -x = Outputs the results in XML format (see [“-x \(Output in XML Format 1\)” on page 5-35](#)). For example:  
# **qauccli -pr fc -i all -x -o output.xml**

You cannot combine the command -f with any other options:

- -f = Input Parameter Options from a Text File (see [“-f \(Input Parameter Options from a Text File\)” on page 5-11](#)). For example:  
# **qauccli -pr fc -f command.txt**

In addition, the following general rules for commands apply:

- Only *one* command line option per input file is valid.
- You can use either the hyphen (-) character or the forward slash (/) character. For example, both of these commands are valid:

```
qauccli -pr fc -g
qauccli -pr fc /g
```

[Table 5-2](#) defines the command variables.

**Table 5-2. Command Variables**

| Variable       | Definition                   | Format                                         |
|----------------|------------------------------|------------------------------------------------|
| <hba instance> | Adapter number <sup>a</sup>  |                                                |
| <hba wwpn>     | Adapter world wide port name | xx-xx-xx-xx-xx-xx-xx-xx or<br>xxxxxxxxxxxxxxxx |
| <alias>        | Adapter alias                | Symbolic adapter <sup>b</sup> name             |
| <target wwnn>  | Target world wide node name  | xx-xx-xx-xx-xx-xx-xx-xx or<br>xxxxxxxxxxxxxxxx |
| <target wwpn>  | Target world wide port name  | xx-xx-xx-xx-xx-xx-xx-xx or<br>xxxxxxxxxxxxxxxx |

**Table 5-2. Command Variables (Continued)**

| Variable         | Definition          | Format             |
|------------------|---------------------|--------------------|
| <target port id> | Target port ID      | xx-xx-xx or xxxxxx |
| <target id>      | Target ID           |                    |
| <lun id>         | Logical unit number | (0–255)            |
| <address>        | IP address          | xxx.xxx.xxx.xxx    |
| ( speed )        | Target link speed   | 1.2, 4, or 8GHz    |

<sup>a</sup> You can use the -g command to find adapter numbers.

<sup>b</sup> You assign the symbolic name. It is limited to 100 characters in length.

## Command Summary

This subsection lists and describes each command line option in alphabetic order, followed by a command description.

### -a

#### (View or Toggle Beacon Status)

Use the -a command to either view (1) or change (2) the status of a beacon.

#### NOTE:

- Adapter must not be in silent mode. See “-s (Silent Mode)” on page 5-31 to issue a response to this command.
- The -a feature is not supported on QLA22xx adapters or QLE8042 adapter.

To view the adapter port's LED beacon state, type:

```
# qaucLi -pr fc -a (<hba instance> | <hba wwpn>)(view | ?)
```

When the adapter's LED is flashing, the following message appears:

```
HBA Port x - LED Flashing is ON.
```

When the adapter's LED is not flashing, the following message appears:

```
HBA Port x - LED Flashing is OFF.
```

To start or stop flashing the adapter's LED beacon, type:

```
# qaucLi -pr fc -a (<hba instance> | <hba wwpn>)
```

The QConvergeConsole CLI toggles the LED's state. If the LED is flashing, the flashing stops. If the LED is not flashing, the flashing begins.

If the QConvergeConsole CLI is not in silent mode (see “-s (Silent Mode)” on page 5-31), one of the following messages appears to indicate the LED's current state:

The LED Flashing for *<hba instance>* *<hba wwpn>* has been turned ON

The LED Flashing for *<hba instance>* *<hba wwpn>* has been turned OFF

The *hba instance* and *hba wwpn* variables must match the command input parameter.

## **-b** **(Save or Update the Flash BIOS or FCode)**

Use the -b command to:

- Update the BIOS or FCode from a file (1).
- Save the BIOS or FCode to a file (2).

To update one or all adapter's Flash with new BIOS or FCode, type:

```
# qauccli -pr fc -b (all | <hba instance> | <hba wwpn>) [(<-rg> | all)] <BIOS/FCode File Name>
```

Where:

all = All adapters of the same type in the system are updated with the new BIOS/FCode

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name

-rg = Flash region update mode (4GB adapters only<sup>a</sup>)

all Update all regions (BIOS/FCode/EFI) depending on the Flash

*BIOS/FCode File Name* = Name or path of file containing update BIOS/FCode

<sup>a</sup> Region update only supported on QLA/QLE/QMC246x adapters.

To save the BIOS/FCode to a file, type:

```
# qauccli -pr fc -b (<hba instance> | <hba wwpn>) SAVE <BIOS/FCode File Name>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*BIOS/FCCode File Name* = File name or a path to a file in which to save BIOS/FCCode

## **-C** **(Show Parameter Settings)**

Use the -c command to show the parameter settings for all adapters in the system:

```
# qaucli -pr fc -c [ <all> ]
```

To show the parameter settings for a specific adapter, type:

```
# qaucli -pr fc -c (<hba instance> | <hba wwpn>)
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter

## **-cna** **(FCoE Utilities Menu for QLx81xx Adapters)**

Use the -cna command to work with converged network adapters (QLE81xx and QMI81xx).

The -cna command line options use the following variables:

*hba instance* = Adapter instance number of a converged network adapter port  
*hba wwpn* = World wide port name of a converged network adapter port  
mpi file = MPI configuration DAT file  
--detail = View detailed format  
--raw = View raw format  
param name = Parameter name  
param alias = Parameter alias  
param value = Parameter value

To view general information for the adapter, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --geninfo
```

To view data center bridging exchange protocol settings, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --dcb --info
```

To update the message passing interface (MPI) configuration table on the adapter from a file, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --updatecfg <mpi file>
```

To save the current MPI configuration table of the selected adapter to a file, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --savecfg <mpi file>
```

To view DCBX Type-Length-Value (TLV) data sent and received over the adapter port connection, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --tlv --detail | --raw
```

To view data center Ethernet (DCE™) network activity information sent and received over the adapter port connection, issue the following command:

```
qauccli -pr fc -cna <hba instance>|<hba wwpn> --mon --dce [ { <param name> | <param alias> <param value> } ]
```

Table 5-3 lists the DCE statistics parameters, possible values, and a description of each.

**Table 5-3. DCE Statistics Parameters**

| Parameter Name | Parameter Alias | Parameter Value | Description                                                                                 |
|----------------|-----------------|-----------------|---------------------------------------------------------------------------------------------|
| AutoPoll       | AP              | 0<br>1–256      | Set to auto polling mode<br>Set to manual polling mode (iteration)                          |
| SetRate        | SR              | 5–30            | Set the polling interval rate (seconds)                                                     |
| Details        | DT              | 0–2             | Set the detailed display mode:<br>■ 0 = Absolute<br>■ 1 = Rate per second<br>■ 2 = Baseline |

## **-dm** **(Display Diagnostics Monitoring Info)**

This option allows you to show general or detailed digital diagnostic monitoring interface for optical transceivers.



**NOTE:**

This option is supported only for QLA/QLE/QEM24xx (4GB) adapters.

To show transceiver monitoring *general* information, type:

```
# qauccli -pr fc -dm (<hba instance> | <hba wwpn> |
<all>)general | gen
```

To show transceiver monitoring *detailed* information, type:

```
# qauccli -pr fc -dm (<hba instance> | <hba wwpn> | all)details
| det
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*all* = All adapters in the system

**-e**

**(Boot Device View | Select | Disable)**

Use the -e command to:

- Show the current boot-device selection for all adapters.
- Show the current boot-device selection for a specific adapter.
- Set a target device as the boot device for an adapter.
- Selectable boot—The OS boots from the first target the BIOS finds.
- Delete the boot device from an adapter.

In these commands:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*target wwnn* = World wide node name of the target

*target wwpn* = World wide port name of the target

*target id* = ID to which the target is bound

*lun id* = ID of the LUN

To view the current boot device selection on all adapters:

```
# qauccli -pr fc -e (view | ?)
```

To view the current boot device selection for a specific adapter:

```
# qauccli -pr fc -e (<hba instance> | <hba wwpn>) (view | ?)
```

In Windows and Linux, each adapter's currently selected boot device is shown in the following format:

```
<hba instance> <hba wwpn> <target wwnn> <target wwpn> <lun id>
```

If the system does not have an adapter with a boot device selected, the following message appears:

```
There is no boot device selected for any HBA in the system.
```

Following is an example of the boot device option when viewing the boot device for a single adapter. In this example, no boot device is configured (selectable boot is disabled; the boot port name is all zeroes).

```
-----  
HBA 1: QLA2340 WWPN xx-xx-xx-xx-xx-xx-xx-xx Port ID 00-00-00  
-----  
Boot Device Settings:  
-----  
Selectable Boot: Disabled  
-----  
(Primary) Boot Port Name LUN  
-----  
00-00-00-00-00-00-00-00 00-00-00-00-00-00-00-00 0  
Press <Enter> to continue:
```

To cause the OS to boot from the first target it finds with a LUN, select a boot port name of 00-00-...00 (selectable boot); this applies only if the boot BIOS is enabled.

To set a specific target as the boot device for a specific adapter, type the following command:

```
# qauccli -pr fc -e (<hba instance> | <hba wwpn>) <target wwnn>  
<target wwpn> <target id> <lun id>
```

**NOTE:**

The boot device setting option is not supported on Macintosh® PowerPC.

The QConvergeConsole CLI checks all parameters to verify that the adapter, targets, and LUNs are valid.

If you select an adapter with no target or a target with no LUN, the QConvergeConsole CLI shows an error message and aborts.

For all operating systems, if the adapter already has boot devices defined and they are different from the ones in the command parameter or menu selection, you are prompted to confirm the boot device selection:

```
The HBA already has a boot device(s) selected. Do you want to
replace it with the new one?
```

For all operating systems, if the adapter already has boot devices selected and they are the same as the ones in the command parameter or menu selection, the following message appears:

```
The HBA already has that device selected as boot device.
```

To use selectable boot (the OS boots from the first target the BIOS finds) set the `<target wwpn>`, `<target wwnn>` and `<lun id>` inputs to all zeroes. For example type:

```
# qauccli -pr fc -e (<hba instance> | <hba wwpn>) (enable | 0 0
0)
```

To disable (delete) the boot device for a specific adapter, type:

```
# qauccli -pr fc -e (<hba instance> | <hba wwpn>) disable
```

## **-ei** (List Fibre Channel Error Codes)

This command lists the Fibre Channel error codes.

## **-f** (Input Parameter Options from a Text File)

### **NOTE:**

The `-f` option is valid only in noninteractive mode, and cannot be combined with any other options. Only one command line parameter per file is valid. This option is used when it is run as a script file.

To input parameter options to the QConvergeConsole CLI through a text file, type `-f`, followed by the file name. For example:

```
# qauccli -pr fc -f command.txt
```

The text file must be formatted as follows:

- The file must contain a single line.
- The file must contain only parameters.
- The file cannot contain another `-f` option.

The following example shows how to set the connection option of an adapter to default (loop preferred, otherwise point-to-point) and the data rate to auto through a command file that is invoked by the `-f` option:

1. Create a text file (for example, `setadapter0.txt`) and enter the following command in the text file:

```
-n 0 co 2 dr 2
```

2. Save and close the file.
3. Issue the `qauccli -pr fc` command with the `-f` option (input from command file):

```
# quaccli -pr fc -f setadapter0.txt
# quaccli -pr fc -f <file name>
```

The QConvergeConsole CLI specifies command line input from file.

## **-fg** (Show Driver Settings)

### **NOTE:**

The `-fg` option is valid only in noninteractive mode.

To show the driver settings, type:

```
# quaccli -pr fc -fg (<hba instance> | <hba wwpn>) (view | ?)
```

Where:

*hba instance* = Adapter number (use `-g` command to find)

*hba wwpn* = World wide port name of the adapter

## **-fs** (Configure Driver Settings)

Type the following command to configure the driver settings:

```
# quaccli -pr fc -fs (<hba instance> | <hba wwpn>) {(<param name> | <param alias>) <param value>}
```

Where:

*hba instance* = Adapter instance number of an adapter port (use `-g` command to find)

*hba wwpn* = World wide port name of the adapter

*param name* = Name of the parameters

*param alias* = Alias of the parameters  
*param value* = New value of the parameters

The pairs *<param name>* *<param value>* and *<param alias>* *<param value>* can be repeated to set multiple parameters in a single command.

Table 5-4 lists the driver configuration parameter names and aliases.

**Table 5-4. Driver Settings Parameters**

| Description                                                                     | Parameter Name<br><param name> | Alias<br><param alias> | Value <sup>a</sup><br><param value> |
|---------------------------------------------------------------------------------|--------------------------------|------------------------|-------------------------------------|
| Persistently bound target(s) only                                               | PersistentOnly                 | PO                     | 0, 1                                |
| Present persistently bound target(s) plus any new target(s) with driver default | PersistentPlusNew              | PN                     | 0, 1                                |
| Present targets with driver default                                             | NewOnly <sup>b</sup>           | NO                     | 0, 1                                |
| Bind devices by WWPN                                                            | BindWWPN                       | BW                     | 0, 1                                |
| Bind devices by port ID                                                         | BindPortID                     | BP                     | 0, 1                                |

Table Notes

<sup>a</sup> 0 = Disabled, 1 = Enabled

<sup>b</sup> Driver parameter NewOnly (NO) is supported only with the failover driver. If you select a QLA22xx adapter, all QLA22xx adapters on the host will use the same settings. If you select a 23xx adapter, all 2G/4G adapters on the host will also use the same settings.

The following restrictions apply:

- Under Linux, this feature is disabled if you are using the IOCTL or sysfs (inbox) driver.
- Under Mac OS®, the BindWWPN and BindPortID parameters are not changeable; Mac OS supports only the BindWWPN parameter.

## -gs (Show HBA Statistics)

To view the adapter statistics, type:

```
# qaucli -pr fc -gs (<hba instance> | <hba wwpn>) [(<param name> | <param alias>) <param value>]
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*param name* = Name of the parameter (see table 4-7)

*param alias* = Alias of the parameter (see table 4-7)

*param value* = New value of the parameter (see table 4-7)

Table 5-5 lists the adapter statistics parameter names, aliases, and values.

**Table 5-5. Adapter Statistics Parameters**

| Description                                                   | Name      | Alias | Value              |
|---------------------------------------------------------------|-----------|-------|--------------------|
| Sets how often statistics are retrieved                       | AutoPoll  | AP    | 0–256 <sup>a</sup> |
| Set the polling interval when retrieving statistics (seconds) | PollRate  | SR    | 5–30               |
| Saves the adapter's statistics to a CVS log file              | LogToFile | LF    | Log file name      |

<sup>a</sup> When the AutoPoll value is 0, statistics are retrieved automatically until the user aborts the operation.

When the AutoPoll value is in the range of 1–256, statistics are retrieved for the number of cycles specified by this value.

Under Linux, this feature is disabled if you are using the IOCTL or sysfs (inbox) driver.

## -h (Help)

Type the following text to view the help file:

```
# qaucLi -pr fc (-h | ?)
```

Type the following text to view help information for an individual command:

```
# qaucLi -pr fc <Command Line Parameter> (-h | ?)
```

For example, typing **qauccli -pr fc -l -h** shows:

```
QConvergeConsole CLI Utility
v1.x.x Build x
Copyright 2003-2009 QLogic Corp.
All rights reserved.
Command Line QLogic FC Host Bus Adapters

qauccli -pr fc -l (<hba no> | <hba wwpn>)
    - shows LUN information for all HBAs.
qauccli -pr fc -l (<hba no> | <hba wwpn>) (<target wwpn>|
<target portid>)
    - shows LUN information for a specific target
qauccli -pr fc -l (<hba no> | <hba wwpn>) (<target wwpn>|
<target portid>) <lun id>      - shows LUN information for a
specific LUN on a specific target
```

#### NOTE:

To view the current version information for the QConvergeConsole CLI, use the -v option described in [“-v \(Display Version\)” on page 5-34](#).

## -ha (Set | Delete HBA Alias)

With this command you can view, set or delete the alias of a specific adapter.

To view the alias of a specified adapter, type:

```
# quaccli -pr fc -ha (<hba instance> | <hba wwpn>) view | ?
```

To set an alias for a specified adapter, type:

```
# quaccli -pr fc -ha (<hba instance> | <hba wwpn>) <alias>
```

To delete an alias of a specified adapter, type:

```
# quaccli -pr fc -ha (<hba instance> | <hba wwpn>) delete
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*alias* = Symbolic adapter name. (100 characters, maximum)

To set an alias for the selected adapter:

```
# quaccli -pr fc -ha (<hba instance> | <hba wwpn>) <alias>
```

To delete the current alias of the selected adapter:

```
# qaucli -pr fc -ha (<hba instance> | <hba wwpn>) delete
```

To view the current alias of the selected adapter:

```
# qaucli -pr fc -ha (<hba instance> | <hba wwpn>) view | ?
```

## **-i**

### **(Display HBA General Information or VPD Information for One or All HBAs)**

To show general adapter information or VPD information for all adapter ports in the system, type the following command:

```
# qaucli -pr fc -i [ <all> ] [ <VPD> ]
```

Where:

*all* = All adapters in the system  
*VPD* = Vital Product Data

To show general or VPD information for a specific adapter, type:

```
# qaucli -pr fc -I (<hba instance> | <hba wwpn>) [ VPD ]
```

To show VPD (QLA/QLE24xx only) for all adapter ports in the system, type:

```
# qaucli -pr fc -i VPD
# qaucli -pr fc -i all VPD
```

Where:

*all* = VPD information for all adapters in the system

## **-kr**

### **(Run Read/Write Buffer Test)**

Use the **-kr** command to do read/write buffer test.

To do a read/write test using the default parameters, type:

```
# qaucli -pr fc -kr (<hba instance> | <hba wwpn>)
```

To do a read/write test with customized parameters, type:

```
# qaucli -pr fc -kr (<hba instance> | <hba wwpn>) [(-EX |
-EXCLUDE) <target wwpn>] (<param name> | <param alias>) <param
value>
```



Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

-ex or -exclude = World wide port name of the target device that is excluded from the read/write test

*param name* = One of the following:

- DataPattern
- DataSize
- TestCount
- TestIncrement
- OnError

*param alias* = One of the following:

- DP
- DS
- TC
- TI
- OE

*param value* = Value for selected parameter

[“Run a Read/Write Buffer Test” on page 5-17](#) contains command line examples and test results for a read/write buffer test.

## Run a Read/Write Buffer Test

This test runs on all devices or on the devices you select on the adapter you select. This test sends the SCSI Write Buffer command to the target devices and uses the SCSI Read Buffer command to read the returned data and do a data comparison.

### NOTE:

All devices attached to the adapter must support SCSI Read/Write Buffer commands.

To do a read/write buffer test using the current parameters, type:

```
# qaucLi -pr fc -kr (<hba instance> | <hba wwpn>)
```

To exclude a device or devices on the selected adapter port, type:

```
# qauccli -pr fc -kr (<hba instance>|<hba wwpn>) [(-EX |
-EXCLUDE) <target wwpn>] [(<param name> | <param alias>)
<param value>]
```

The following examples do 500 read/write buffer tests with a test increment of 10, a data pattern of FFh, and a data size of 8 bytes. The test stops if an error occurs.

```
# qauccli -pr fc -kr <hba instance> DP FF DS 8 TC 500 TI 10 OE 1
# qauccli -pr fc -kr <hba wwpn> DP FF DS 8 TC 500 TI 10 OE 1
```

You do not have to set each parameter for the read/write buffer test. If you do not specify a parameter in the command line, the current parameter value is used. For example, the following command sets the data pattern to FFh and uses the current values for the other parameters:

```
# qauccli -pr fc -kr <hba instance> DP FF
```

The following example sets the data size to 16 bytes and has the test loop if an error occurs. The other parameters use the current values.

```
# qauccli -pr fc -kr <hba wwpn> DS 16 OE 2
```

An example of a successful read/write buffer test follows:

HBA 1: QEM2462 Port 1 WWPN xx-xx-xx-xx-xx-xx-xx-xx Port ID 02-0E-00

-----  
 Test Configuration  
 -----

```
Data Pattern           : Random
Data Size (Bytes)      : 512
Number of test(s) (0-65535) : 500
Test Increment (1-65535) : 1
On Error               : Ignore
Test Continuous        : OFF
```

### Run Adapter Diagnostics Read-write Buffer Test

The {(-ex | exclude) <target wwpn>} parameters specify a device that will be excluded from the read/write buffer test.

The system shows the following information after a read/write buffer test completes:

- Loop ID/status
- Data miscompare
- Link failure
- Loss of sync

- Loss of signal
- Invalid CRC

Table 5-6 lists the *<param name>*, *<param alias>*, and *<Param value>* options.

**Table 5-6. Diagnostics Parameters**

| Name<br><param name>       | Alias<br><param alias> | Value<br><param value>                                                                                                                             | Description                                                                                                        |
|----------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| DataPattern                | DP                     | Test pattern in hex format <sup>a</sup><br>Customized (00–FF)<br>Random pattern<br>CRPAT <sup>b</sup><br>CJTPAT <sup>c</sup><br>CSPAT <sup>d</sup> | 00, 55, 5A, A5, AA, FF (see Table 5-7)<br>—<br>—<br>Loopback test only<br>Loopback test only<br>Loopback test only |
| DataSize <sup>e</sup>      | DS                     | 8, 16, 32, 64, 128, 256, 512, 1024, 2048                                                                                                           | Actual data transferred during any given pass of the test                                                          |
| TestCount <sup>f</sup>     | TC                     | 0–65535<br>0–10,000                                                                                                                                | Loopback test only<br>Read/write buffer test only                                                                  |
| TestIncrement <sup>g</sup> | TI                     | 1–65535<br>1–10,000                                                                                                                                | Loopback test only<br>Read/write buffer test only                                                                  |
| OnError                    | OE                     | 0–2                                                                                                                                                | 0 = ignore<br>1 = stop<br>2 = loop on error                                                                        |

<sup>a</sup> Valid two-character case-insensitive hexadecimal patterns.

<sup>b</sup> Compliant random data pattern in a valid Fibre Channel frame, as defined by the ANSI document *Methodologies for Jitter and Signal Quality Specification—MJSQ Annex A —Test bit sequences*.

<sup>c</sup> Compliant jitter tolerance pattern in a Fibre Channel frame, as defined by the ASIC document listed above.

<sup>d</sup> Compliant supply noise test sequence in a valid Fibre Channel frame, as defined by the ASIC document listed above.

<sup>e</sup> For read/write buffer test, the maximum size is 128; this is also the default.

<sup>f</sup> 0=test continuously. 1–10,000 and 1–65535=total number of tests that will be executed.

<sup>g</sup> Must be less than the TestCount value.

**Table 5-7. Data Pattern (DP) Test Patterns**

| Hex | Binary   |
|-----|----------|
| 00  | 00000000 |
| 55  | 01010101 |
| 5A  | 01011010 |
| A5  | 10100101 |
| AA  | 10101010 |
| FF  | 11111111 |

If the read/write buffer test fails, the system shows the following information:

- **Loop/port ID** (the loop ID of the adapter when operating in loop mode).
- **Status:**
  - ☐ **Success**—The test passed.
  - ☐ **Error**—A data miscompare or link status firmware error occurred.
  - ☐ **Failed**—A link status error, SCSI write buffer error, or SCSI read buffer error occurred.
  - ☐ **Unknown**—The target was not present.
  - ☐ **Unsupported**—The device does not support this test.
- **Data Miscompare**—The possible values are:
  - ☐ 0 (no data miscompares)
  - ☐ Device not present
  - ☐ Get link status failed
  - ☐ Read buffer failed
  - ☐ R/W buffer not supported
  - ☐ Write buffer failed
- **Link Failure**—Number of link failures
- **Loss of sync**—Number of sync loss errors
- **Loss of signal**—Number of signal loss errors
- **Invalid CRC**—Number of invalid CRCs

## -l (Display LUN Info)

This command shows LUN information for:

- All adapters
- A specific target
- A specific LUN on a specific target

To show the LUN information for *all target* devices for a *specific adapter Instance*, type:

```
# qauccli -pr fc -l (<hba instance> | <hba wwpn>)
```

To show the LUN information for a specific device for a specific adapter port, type:

```
# qauccli -pr fc -l (<hba instance> | <hba wwpn>) (<target port id> | <target wwpn>)
```

To show the LUN information for a *specific LUN* on a *specific target* device for a *specific adapter* port instance, type:

```
# qauccli -pr fc -l (<hba instance> | <hba wwpn>) (<target port id> | <target wwpn>) <lun id>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*target port id* = Port ID of the target  
*target wwpn* = World wide port name of the target  
*lun id* = ID of the LUN

## -ls (Display Link Status)

To view the link status, type:

```
# qauccli -pr fc -ls (<hba instance> | <hba wwpn>) [(<param name> | <param alias>) <param value>]
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*param name* = Name of the parameter (see [Table 5-8](#))  
*param alias* = Alias of the parameter (see [Table 5-8](#))  
*param value* = New value of the parameter (see [Table 5-8](#))

Table 5-8 defines the link status parameter names, aliases, and values.

**Table 5-8. Link Status Parameters**

| Description                                 | Name      | Alias | Value                     |
|---------------------------------------------|-----------|-------|---------------------------|
| Sets link-status retrieval period           | AutoPoll  | AP    | 0–256 <sup>a</sup> cycles |
| Sets link-status retrieval polling interval | PollRate  | SR    | 5–30 seconds              |
| Save link status to CVS log file            | LogToFile | LF    | Log file name             |

<sup>a</sup> When the AutoPoll parameter is 0, the link status is retrieved automatically until the user aborts the operation. When the AutoPoll parameter is in the range of 1–256, the link status is retrieved for the number of cycles specified by this value.

## **-m** (View, Enable, or Disable LUNs)

### **NOTE:**

- If the current driver setting is Bind by Port ID, this option is not available.
- Under Linux, if you use the IOCTL or sysfs (inbox) driver, this feature is disabled.
- You must persistently bind the targets before configuring selective LUNs, (link).

Use the `-m` command to:

- View all selective LUNs for all adapter ports
- View an adapter's selective LUN list
- View the current selective state of a LUN on a specific target
- Enable (select) a LUN on a specific target on a specific adapter
- Disable (deselect) a LUN on a specific target on a specific adapter
- Enable all LUNs on a specific target on a specific adapter
- Disable (deselect) all LUNs on a specific target on a specific adapter
- Enable (select) all LUNs of all targets on a specific adapter
- Disable (deselect) all LUNs of all targets on a specific adapter

For these commands:

*hba instance* = Adapter number (use `-g` command to find)  
*hba wwpn* = World wide port name of the adapter  
*target wwnn* = World wide node name of the target  
*target wwpn* = World wide port name of the target  
*lun id* = ID of the LUN

To view all selective LUNs for all adapter ports, type:

```
# qauccli -pr fc -m all (view | ?)
```

To view an adapter's selective LUN list, type:

```
# qauccli -pr fc -m (<hba instance> | <hba wwpn> | all) (view  
| ?)
```

The QConvergeConsole CLI shows the adapter's enabled LUN list in the following format:

```
<target wwnn> <target wwpn> <lun id>
```

To view the current select state of a specific LUN, type:

```
# qauccli -pr fc -m (<hba instance> | <hba wwpn>) (<target  
wwnn> <target wwpn> <lun id>) (view | ?)
```

If the input represents a valid LUN, the QConvergeConsole CLI shows that LUN's current state as selected or deselected for that adapter.

To enable (select) a LUN on a specific target on a specific adapter, type:

```
# qauccli -pr fc -m (<hba instance> | <hba wwpn>) {<target  
wwnn> <target wwpn> <lun id> ( 1 | enable | select )}
```

**NOTE:**

You can repeat the sequence `<target wwnn> <target wwpn> <lun id> 1` to select multiple LUNs in the same command.

To disable (deselect) a LUN on a specific target on a specific adapter, type:

```
# qauccli -pr fc -m (<hba instance> | <hba wwpn>) {<target  
wwnn> <target wwpn> <lun id> (0 | disable | deselect)}
```

**NOTE:**

To select multiple LUNs in the same command, repeat the following sequence:

```
<target wwnn> <target wwpn> <lun id> 0
```

To enable (select) all LUNs for a specific target on a specific adapter, type:

```
# qauccli -pr fc -m (<hba instance> | <hba wwpn>) select |  
enable <target wwnn> <target wwpn>
```

To disable (deselect) all LUNs for a specific target on a specific adapter, type:

```
# qaucli -pr fc -m (<hba instance> | <hba wwpn>) deselect |
disable <target wwn> <target wwpn>
```

To enable (select) all LUNs of all targets on a specific adapter:

```
# qaucli -pr fc -m (<hba instance> | <hba wwpn>) select all
```

To disable (deselect) all LUNs of all targets on a specific adapter:

```
# qaucli -pr fc -m (<hba instance> | <hba wwpn>) deselect all
```

## -n

### (Set Selected HBA Parameter)

To set a specific parameter on a specific adapter, type:

```
# qaucli -pr fc -n (<hba instance> | <hba wwpn>) { (<param
name> | <param alias>) <param value> }
```

To restore the default BIOS (4G adapters only), type:

```
# qaucli -pr fc -n (<hba instance> | <hba wwpn>) default
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*param name* = Name of the parameters

*param alias* = Alias of the parameters

*param value* = New value of the parameters

The pairs *<param name>* *<param value>* and *<param alias>* *<param value>* can be repeated to set multiple parameters with a single command.

Table 5-9 lists the adapter parameter names and aliases.

**Table 5-9. Adapter Parameters**

| Description             | Name                                    | Alias | Value                       |
|-------------------------|-----------------------------------------|-------|-----------------------------|
| Connection options      | ConnectionOption                        | CO    | See table note <sup>a</sup> |
| Data rate               | DataRate                                | DR    | See table note <sup>b</sup> |
| Enable BIOS             | EnableBIOS <sup>c</sup>                 | EB    | 1=Enable, 0=Disable         |
| Enable extended logging | EnableExtendedLog-<br>ging <sup>d</sup> | EL    | 1=Enable, 0=Disable         |



**Table 5-9. Adapter Parameters (Continued)**

| Description               | Name                 | Alias | Value                       |
|---------------------------|----------------------|-------|-----------------------------|
| Enable Fibre Channel tape | EnableFCTape         | EF    | 1=Enable, 0=Disable         |
| Enable hard loop ID       | EnableHardLoopID     | HL    | 1=Enable, 0=Disable         |
| Enable LIP full login     | EnableLIPFullLogin   | FL    | 1=Enable, 0=Disable         |
| Enable LIP reset          | EnableLipReset       | LP    | 1=Enable, 0=Disable         |
| Enable target reset       | EnableTargetReset    | TR    | 1=Enable, 0=Disable         |
| Execution throttle        | ExecutionThrottle    | ET    | 1–65535                     |
| Frame size                | FrameSize            | FR    | 512, 1024, 2048             |
| Hard loop ID              | HardLoopID           | HD    | 0–125                       |
| Interrupt delay timer     | InterruptDelay-Timer | ID    | 0–255                       |
| Login retry count         | LoginRetryCount      | LR    | 0–255                       |
| Link down timeout         | LinkDownTimeOut      | LT    | 0–240                       |
| Maximum LUNs per target   | MaximumLUNsPerTarget | ML    | 0, 8, 16, 32, 64, 128, 256  |
| Operation mode            | OperationMode        | OM    | See table note <sup>a</sup> |
| Port down retry count     | PortDownRetryCount   | PD    | 0–255                       |
| Reset Delay               | ResetDelay           | RD    | 0–255                       |

<sup>a</sup> Connection Options:

- 0 = Loop only
- 1 = Point-to-point only
- 2 = Loop preferred, otherwise point-to-point
- 3 = Point-to-point preferred, otherwise loop (QLA22xx adapters only)

<sup>b</sup> Data Rate (QLA23xx/QLA24xx/QLE23xx/QLE24xx and 8Gb adapters):

- 0 = 1 Gb
- 1 = 2Gbs
- 2 = Auto
- 3 = 4 Gbs
- 4 = 8 Gbs

<sup>c</sup> EnableBIOS option not available on 4Gb adapter.

<sup>d</sup> EnableExtendedLogging option not available on PPC64 or SPARC.

- <sup>e</sup> Operation mode (QLA/QLE23xx, QLA/QLE24xx, and 8Gb adapters):
- 0 = interrupt for every I/O completion
  - 5 = interrupt when interrupt delay timer expires
  - 6 = interrupt when interrupt delay timer expires or no active I/O

**NOTE:**

QLA200/210 adapters have a limited set of parameters that can be changed.

## **-O**

### **(Output to a File)**

**NOTE:**

- The `-o` option is valid only in noninteractive mode.
- This option can be used with all noninteractive mode options that have a corresponding interactive mode option (see [Table 5-1](#)). The option must be the first or last command in the command line.
- If the file already exists, new data are appended to the current file.

To output result and status messages into a file, type `-o`, followed by the file name. For example, to save LUN information to a file named `systemLUNinfo`:

```
# qaucLi -pr fc -l -o systemLUNinfo
```

Where the file name is `systemLUNinfo`, all messages are located in the directory indicated for the system platform:

Windows: `syslog.log` in the current directory

Linux: `/var/log/messages`

Mac OS: `/var/log/system.log`

## **-p**

### **(Display Persistent Target Binding Info)**

With this command you can:

- Show binding information for one or on all adapters.
- Bind a specific target to a selected adapter.
- Bind all targets on a specific adapter or on all adapter.
- Unbind a specific target.
- Unbind all targets on a specific adapter or on all adapters.

**NOTE:**

Under Linux, if you use the IOCTL or sysfs (inbox) driver these features are disabled.

To show target persistent binding information for a specific adapter port, type:

```
# qauccli -pr fc -p (<hba instance> | <hba wwpn>) (view | ?)
```

To show persistent binding information for all adapters, type:

```
# qauccli -pr fc -p all (view | ?)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

The following example is a typical QConvergeConsole CLI output showing all targets currently bound to an adapter:

```
-----
HBA 0: QLA2462 Port 1 WWPN xx-xx-xx-xx-xx-xx-xx-xx Port ID 11-06-EF
-----
Bind  Type  Device Node Name          Device Port Name          Port ID  ID
----  -
No    Disk  xx-xx-xx-xx-xx-xx-xx-xx  xx-xx-xx-xx-xx-xx-xx-xx  10-02-E1
Yes   Disk  xx-xx-xx-xx-xx-xx-xx-xx  xx-xx-xx-xx-xx-xx-xx-xx  10-02-E2  0
Yes   Disk  xx-xx-xx-xx-xx-xx-xx-xx  xx-xx-xx-xx-xx-xx-xx-xx  10-02-E4  1
Yes   Disk  xx-xx-xx-xx-xx-xx-xx-xx  xx-xx-xx-xx-xx-xx-xx-xx  10-02-E8  2
Press <Enter> to continue:
```

To bind a selected target to a specific adapter, type:

```
# qauccli -pr fc -p (<hba instance> | <hba wwpn>) (<target
wwnn> <target wwpn> <target port id> <target id>)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwnn* = Word wide node name of the adapter

*hba wwpn* = World wide port name of the adapter

*target wwnn* = World wide node name of the target

*target wwpn* = World wide port name of the target

*target port id* = Port ID of the target

*target id* = ID to which the target is bound

To bind multiple targets with a single command, repeat the following group:

```
<target wwnn> <target wwpn> <target port id> <target id>
```

To bind all targets on a specific adapter or to bind all targets on all adapters, type:

```
# qauccli -pr fc -p (<hba instance> | <hba wwpn> | all) bind
all
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

To unbind a specific target, type:

```
# qauccli -pr fc -p (<hba instance> | <hba wwpn>) (remove
<target wwnn> | unbind <target wwnn>)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*target wwnn* = World wide node name of the target

To unbind all targets on a specific adapter port or on all adapter ports, type:

```
# qauccli -pr fc -p (<hba instance> | <hba wwpn> | all) (remove all
|
unbind all)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

## **-pa** (Define Port Alias)

With this command you can:

- Define the port alias for a specific adapter port (1)
- Delete the port alias from a specific adapter port (2)
- View the port alias for a specific adapter port (3)

To define a port alias for the specified adapter, type:

```
# qauccli -pr fc -pa (<hba instance> | <hba wwpn>) <alias>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*alias* = Symbolic name you assign to the adapter port

To delete a port alias for the specified adapter port, type:

```
# qaucLi -pr fc -pa (<hba instance> | <hba wwpn>) delete
```

To view a port alias for the specified adapter port, type:

```
# qaucLi -pr fc -pa (<hba instance> | <hba wwpn>) view | ?
```

## **-q** (View or Set Target Link Speed)

### **NOTE:**

The -q option is supported only on 4Gb adapters.

Type the following command to view the link speed of all targets attached to one adapter or all adapters:

```
# qaucLi -pr fc -q (<hba instance> | <hba wwpn> | <all>)  
[-targets | -t ]
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*all* = Reports link speed for all targets on all adapters in the system  
*-targets* = All targets  
*-t* = All targets

To view the link speed of a specific target attached to an adapter, enter the following command:

```
# qaucLi -pr fc -q (<hba instance> | <hba wwpn> | <target  
wwpn>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*target wwpn* = World wide port name of the target

To set the link speed of all targets attached to one adapter or all adapters to the designated speed, enter the following command:

```
# qauccli -pr fc -q (<hba instance> | <hba wwpn> | <all>)
<-targets | -t ><speed>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*all* = Reports link speed for all targets on all adapters in the system  
*-targets* = All targets  
*-t* = All targets  
*speed* = Link speed (

To set the link speed of selected target(s) attached to one adapter to the designated speed, type the following command:

```
# qauccli -pr fc -q (<hba instance> | <hba wwpn> | <all>)
<target wwpn> <speed>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*all* = Reports link speed for all targets on all adapters in the system  
*target wwpn* = World wide port name of the target  
*speed* = Link speed

## **-r** **(Update HBA Parameters)**

To update the adapter parameters, type the following:

```
# qauccli -pr fc -r (<hba instance> | <hba wwpn> | all) <File
Name>
```

Where:

*hba instance* = Adapter number (use -g command to find)  
*hba wwpn* = World wide port name of the adapter  
*all* = All adapter ports in the system are updated with the new adapter parameters  
*hba parameters file* = File name or a path to a file that contains the updated adapter parameters  
*name*

To save the adapter parameters to a file, type:

```
# qauccli -pr fc -r (<hba instance> | <hba wwpn>) save <File Name>
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*hba parameters file name* = File name or a path to a file to save the adapter parameters

## **-S** **(Silent Mode)**

### **NOTE:**

- The -s option is valid only in noninteractive mode.
- You can use this option with all noninteractive mode options that have a corresponding interactive mode option (see [Table 5-1](#)). This option must be the first or last command in the command line.

In noninteractive mode, the system shows result and status messages generated by the QConvergeConsole CLI (by default) unless suppressed by silent mode. You can combine this option with the -o option (see ["-o \(Output to a File\)" on page 5-26](#)) so the output is saved in a file and does not appear in the CLI. For example:

```
# qauccli -pr fc -I all -s -o output.txt
```

## **-sp** (Update Adapter Firmware)

To update the adapter firmware SerDes table with a file, type:

```
# qaucli -pr fc -sp <hba instance> | <hba wwpn> <file name>
```

To save the adapter firmware SerDes table to a file, type:

```
# qaucli -pr fc -sp <hba instance> | <hba wwpn> save  
<file name>
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*file name* = Name of the firmware preload table DAT file.

This feature is supported with 8Gb mezzanine/blade adapters. It is not needed with the standard 8Gb adapters.

## **-t** (Display Target Information)

To show the target information for *all* adapters in the system, type:

```
# qaucli -pr fc -t [ <all> ]
```

To show the target information for a *specific* adapter, type:

```
# qaucli -pr fc -t (<hba instance> | <hba wwpn>)
```

To show specific target information for a *specific target* on an adapter, type:

```
# qaucli -pr fc -t (<hba instance> | <hba wwpn>) (<target  
wwpn> | <target port id>)
```

Where:

*hba instance* = Adapter number (use -g command to find)

*hba wwpn* = World wide port name of the adapter

*target wwpn* = World wide port name of the target

*target port id* = Port ID of the target

To show the target information on all adapters, type:

```
-t [<all>]
```

To show the target information on a specific adapter, type:

```
-t (<hba instance> | <hba wwpn>)
```



To show the target information for a specific target on a specific adapter, type:

```
-t (<hba instance> | <hba wwpn>) (<target wwpn> | <target port id>)
```

iiDMA (intelligent interleaved direct memory access) settings include:

- -targets | -t applies to all targets.
- <speed> indicates the supported intelligent interleave factor: 1, 2, 4, or 8Gbps.

## **-tb** (Target Beacon On/Off)

To start or stop flashing a target's LED, type:

```
# qauccli -pr fc -tb (<hba instance> | <hba wwpn>) (<target wwpn>)  
(<0 | ON> | <1 | PRESET>)
```

### **NOTE:**

- The <0 | ON> option flashes the LED until interrupted.  
The <1 | PRESET> option flashes the LED 12 times.
- This feature is supported only on JBOD ("just a bunch of disks") devices.

## **-tp** (Display Host Topology)

Use the -tp option to show the topology configuration of the host:

```
# qauccli -pr fc -tp | topology
```

Under Linux, this feature is disabled if you are using the IOCTL or sysfs (inbox) driver.

## **-u** (Firmware Area Update/Save)

Use the -u option to update the firmware preload area of the adapter from a DAT file or to save the current firmware preload area of the adapter to a DAT file.

### **NOTE:**

This option is available only for 8Gb adapters.

To update the adapter firmware preload table, enter the following command:

```
# qauccli -pr fc -u ( <hba instance> | <hba wwpn> ) <file name>
```

To save the current adapter firmware preload table to a DAT file, enter the following command:

```
# qauccli -pr fc -u ( <hba instance> | <hba wwpn> ) save <file name>
```

Where:

- <hba instance> is the adapter instance number of an adapter port.
- <hba wwpn> is the world wide port name of an adapter port.
- <file name> is the name of the firmware preload table DAT file.

## **-V** (Display Version)

### **NOTE:**

This option is valid only in noninteractive mode.

To show the version number of the QConvergeConsole CLI tool, type:

```
# qauccli -pr fc -v
```

The system shows the following information:

```
QConvergeConsole CLI
v1.x.x Build x
Copyright 2003-2009 QLogic Corp.
All rights reserved.
Command Line QLogic FC Host Bus Adapters.
Build Type: Release
Build Date: xx/xx/xxxx xx:xx:xx AM
```

## **-vp** (Virtual Port)

To list all virtual ports on a physical adapter port, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) list all
```

Where:

- hba instance* = Adapter number (use -g command to find)
- hba wwpn* = World wide port name of the adapter
- vport wwpn* = World wide port name of the virtual port
- vport hex* = World wide port name of the virtual port with the two hex digits in byte three supplied by the user

To list a specific virtual port on a physical adapter port, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) list <vport wwpn>
```

To create a virtual port with an automatic WWPN, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) create auto
```

To create a virtual port with a specific WWPN, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) create  
<vport hex>
```

When prompted, enter two hexadecimal digits. The system checks these digits to be sure they are unique and, if they are, puts them into byte 1 of the WWPN.

To delete all virtual ports on a physical adapter port, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) delete all
```

To delete a specific virtual port on a physical adapter port, type:

```
# qauccli -pr fc -vp (<hba instance> : <hba wwpn>) delete  
<vport wwpn>
```

## **-X** (Output in XML Format 1)

### **NOTE:**

This option is valid only in noninteractive mode.

You can use this option with all noninteractive mode options that have a corresponding interactive mode option (see [Table 5-1](#)). This option must be the first or last command in the command line.

When you use this option, the system shows all result and status messages in XML format 1, a legacy format. This option is usually combined with the -o option (see “[-o \(Output to a File\)](#)” on [page 5-26](#)) to create a text file with XML output so that it can be parsed by an XML-compliant utility. For example to show adapter general information and output it to an XML file named `output.xml`:

```
# qauccli -pr fc -i all -x -o output.xml
```

## **-x2** (Output in XML Format 2)

**NOTE:**

This option is valid only in noninteractive mode.

You can use this option with all noninteractive mode options that have a corresponding interactive mode option (see [Table 5-1](#)). This option must be the first or last command in the command line.

When you use this option, the system shows all result and status messages in XML format 2, standard XML format. This option is usually combined with the `-o` option (see [“-o \(Output to a File\)” on page 5-26](#)) to create a text file with XML output so that it can be parsed by an XML-compliant utility. For example, to show adapter general information and output it to an XML file named `output.xml`:

```
# qaucli -pr fc -i all -x2 -o output.xml
```

## **-Z** (Display All HBA information for One or All HBAs)

To show all information for one specific adapter or for all adapters in the system, type:

```
# qaucli -pr fc -z (<hba instance> | <hba wwpn>) | <all>
```

[“Host Configuration \(Command Line Option -z\)” on page 5-2](#) covers this command.

# 6 iSCSI Noninteractive Commands

This section describes the noninteractive iSCSI command format and parameters of the QConvergeConsole CLI for the QLogic 8242 Converged Network Adapter.

**NOTE:**

To view help, enter the command **-h**.

## Command Format

The noninteractive iSCSI command format is as follows:

```
gauccli -pr iscsi [optional parameters] <Command Option> [Command Variable] | [Command Variable]
```

Where:

<> = Parameters in angled brackets are required.

[] = Parameters in square brackets are optional.

| = OR operator.

*Italicized text* = Placeholder describing the command variable.

Table 6-1 defines the command variables.

**Table 6-1. Noninteractive Command Variables**

| Variable             | Definition                  |
|----------------------|-----------------------------|
| <i>hba_port_inst</i> | System port                 |
| <i>Target ID</i>     | Target ID                   |
| <i>LUN ID</i>        | Logical unit number (0–255) |

**Table 6-1. Noninteractive Command Variables (Continued)**

| Variable           | Definition                                                                                          |
|--------------------|-----------------------------------------------------------------------------------------------------|
| <i>CHAP Number</i> | Challenge handshake authentication protocol (CHAP) number                                           |
| <i>CHAP Name</i>   | Null-terminated CHAP name, which is sent by the port instance when responding to the CHAP challenge |
| <i>CHAP Secret</i> | CHAP secret, which is used by the port instance when generating the CHAP response                   |
| <i>Boot Target</i> | The target ID of the target device from which to boot                                               |
| <i>Boot LUN</i>    | The LUN of the boot device                                                                          |

**NOTE:**

- The commands are case sensitive. For example, -g is not the same as -G.
- Only one command line option per input file is valid.
- *<hba\_port\_inst>* is an instance number from the list of all iSCSI adapter ports in the system. You can see this list with the command -i.

The positional parameters vary, depending on the command option specified. For example, to display an adapter's configured settings, use the command option -C, followed by the port instance number (positional parameter). In contrast, the command for providing general information, -g, has no positional parameters.

In noninteractive mode, most changes made to the port instance become effective immediately. This sometimes causes the adapter to reset, as specified in the command description. Be sure the system is prepared for an adapter reset before making changes.

This remainder of this chapter contains an alphabetical list of the QConvergeConsole CLI commands used in noninteractive mode.

## Command Summary

This subsection lists and describes each command line option in alphabetic order, followed by a command description.

## **-acb**

To inquire whether the access method control block (ACB) firmware functions are supported, issue the `-acb` command. In general, up-to-date firmware and driver are required for ACB to be supported.

## **-addchap** (Add a CHAP Entry)

To add a CHAP entry to the persistent CHAP table, issue the `-addchap` command as follows:

```
-addchap [ -BIDI ] <hba_port_inst> <CHAP Name> <CHAP Secret>
```

The optional parameter `[-BIDI]` shows that the CHAP entry is BIDI (bidirectional). When this parameter is not specified (default), the CHAP entry is local.

Issuing this command resets the adapter.

## **-arp** (Display ARP Table)

The address resolution protocol (ARP) cache keeps a record of host port connections with other hardware (such as targets) on the network. The IP address/MAC address pairs are dynamic entries that are removed after 10 minutes. To view the ARP table, issue the `-arp` command as follows:

```
-arp <hba_port_inst>
```

## **-b** (HBA Reset)

To reset the specified adapter after making configuration changes, issue the `-b` command as follows:

```
-b <hba_port_inst>
```

### **NOTE:**

- Issuing this command resets both ports (0 and 1) on two-port adapters.
- You must disable boot-code mode to reset the adapter (see [“-bootcodemode \(Set BIOS/UEFI \[or FCode\] Mode\)” on page 6-4](#)).

## **-binfo** (Display BIOS/UEFI [or FCode] Information)

To display the boot-code settings for the specified adapter port, issue the `-binfo` command as follows:

```
-binfo <hba_port_inst>
```

## **-bootcode** (Update BIOS/UEFI [or FCode] Mode)

Boot code (that is, the bootable code image) is code that allows system boot from an iSCSI drive.

The -bootcode command updates the boot-code image, which should be done when QLogic releases a new boot code with bug fixes or enhancements. At the prompt, type the name of the file containing the boot-code image to upload to the adapter.

To update the boot-code image, issue the -bootcode command as follows:

```
-bootcode <hba_port_inst> <Bootcode Code Image File Name>
```

## **-bootcodemode** (Set BIOS/UEFI [or FCode] Mode)

To set the boot-code mode for QLA4050, QLA4050C, QLE4060C, QLE4062, and QMC4052 adapters, issue the -bootcodemode command as follows:

```
-bootcodemode <hba_port_inst> <1>|<2>|<3>
```

Where:

- 1 = Disabled
- 2 = Manual mode
- 3 = DHCP-Root path

If adapter DHCP is enabled (see ["-ipdhcp \(Configure IP Settings\)" on page 6-13](#)), modes 1–3 are allowed. If adapter DHCP is not enabled, boot-code DHCP is not allowed. This mode is not supported for QLA4010 adapters. This mode is not active for the PPC.

For IBM cards, the following modes are available:

- 4 = DHCP vendor ID
- 5 = DHCP-auto
- 6 = Auto
- 7 = System mode

### **NOTE:**

Boot-code DHCP is not allowed for QLA4010 adapters.



## **-c** (Display Configured Port Settings)

To view the configured settings for the port, issue the -c command as follows:

```
-c [hba_port_inst]
```

When you specify a port instance, information for only that adapter is shown.  
When you do not specify a port instance, information for all adapter ports in the system is shown.

## **-ch** (HBA Information)

To display the adapter's configured settings, issue the -ch command as follows:

```
-ch [hba_port_inst]
```

When you specify a port instance, information for only that adapter is shown.  
When you do not specify a port instance, information for all adapter ports in the system is shown.

## **-chapmap** (Display Targets Using CHAP Entries)

To display the map of targets to CHAP table entries, issue the -chapmap command as follows:

```
-chapmap <hba_port_inst>
```

## **-cpbootcode** (Clear Primary Boot Target Information)

To clear the primary boot target and LUN, issue the -cpbootcode command as follows:

```
-cpbootcode <hba_port_inst>
```

### **NOTE:**

This command is not active for the PPC.

## **-csbootcode** (Clear Secondary Boot Target Information)

To clear the secondary boot target and LUN, issue the -csbootcode command as follows:

```
-csbootcode <hba_port_inst>
```

## **-d** (Install HBA Driver, All Adapters)

To install an adapter driver for all iSCSI adapters on the system from a .zip or .inf file, issue the -d command as follows:

```
-d <file_name>
```

A system reboot may be required; see the log file for details.

## **-dc** (Display Destination Cache [IPv6 only])

The IPv6 destination cache contains the IP address, next-hop IP address, and path MTU information about both local and remote destinations. For multicast and on-link unicast destinations, the next-hop IP address always matches the destination IP address. For unicast destinations that are off-link, the next-hop IP address is the IP address of the router.

To display the destination cache, issue the -dc command as follows:

```
-dc <hba_port_inst>
```

## **-dce**

To display the connection error logs for up to 100 latest connections, issue the -dce command as follows:

```
-dce <hba_port_inst> <target_id> <filename>
```

If <hba\_port\_inst> and <target\_id> are not specified, all entries for all adapter ports are displayed. If <filename> is specified, then the log is written to that file.

## **-delchap** (Delete a CHAP Entry)

To delete the specified CHAP entry, issue the -delchap command as follows:

```
-delchap <hba_port_inst> <CHAP>
```

Issuing this command resets the adapter.

## **-df**

To view the firmware properties, issue the -df command as follows:

```
-df <hba_port_inst>
```

The QConvergeConsole CLI displays information about the adapter: model, serial number, port number, iSCSI name, alias, IP address, instance number, and the following firmware information:

- FW information for instance

- FW version
- FW attribute
- FW version 2
- FW attribute 2

## **-dp**

Use the `-dp` command to preinstall an adapter driver specified from a `.zip` or `.inf` file. This operation may be performed without an adapter installed in the system.

You can specify the following actions:

- **p** — Preinstall driver package (default)
- **i** — Install driver package
- **u** — Uninstall driver package
- **g** — Get driver package path

You may also specify the following flags:

- `DRIVER_PACKAGE_FORCE`
- `DRIVER_PACKAGE_LEGACY_MODE`
- `DRIVER_PACKAGE_ONLY_IF_DEVICE_PRESENT`
- `DRIVER_PACKAGE_REPAIR`
- `DRIVER_PACKAGE_SILENT`

You may need to reboot the system following this command. For details about the success or failure of the driver update operation, view the trace log file.

## **-dr**

### **(Display Default Router List [IPv6 only])**

The IPv6 default router list includes both the active default router and a short list of other routers that advertised themselves as capable of acting as a default router. The active default router always appears first in the list.

To view the default router list, issue the `-dr` command as follows:

```
-dr <hba_port_inst>
```

## **-dspchap**

### **(Display a CHAP Table)**

To view the CHAP table, issue the `-dspchap` command as follows:

```
-dspchap <hba_port_inst>
```

## **-dtdsp** (Display Discovered Targets)

To view the discovered targets that are not persistent, issue the `-dtdsp` command as follows:

```
-dtdsp <hba_port_inst>
```

## **-dtdspa** (List Targets)

To view discovered targets, including both persistent and nonpersistent, issue the `-dtdspa` command as follows:

```
-dtdspa <hba_port_inst>
```

If the specified port instance has no targets assigned, the CLI returns an error.

## **-dtdupd** (Duplicate a Discovered Target)

To duplicate a discovered target that is not persistent, issue the `-dtdupd` command as follows:

```
-dtdupd <hba_port_inst> <target_instance>
```

To obtain the target instance, use the command [“-dtdsp \(Display Discovered Targets\)” on page 6-8](#).

A duplicate target is assigned a new iSCSI initiator ID (ISID) and can then be used to create a redundant path.

## **-dtdupd** (Duplicate a Persistent Target)

To duplicate a discovered target that is persistent, issue the `-dtdupd` command as follows:

```
-dtdupd <hba_port_inst> <target_id>
```

A duplicate target is assigned a new iSCSI initiator ID (ISID) and can then be used to create a redundant path.

## **-dtli** (Login and Persist a Discovered Target)

To log in a specific discovered target, issue the `-dtli` command as follows:

```
-dtli <hba_port_inst> <target_instance>
```

To obtain the target instance, use the command [“-dtdsp \(Display Discovered Targets\)” on page 6-8](#).

## **-dtlia** (Login and Persist a Discovered Target)

To log in all discovered targets, issue the `-dtlia` command as follows:

```
-dtlia <hba_port_inst>
```

## **-dtrem** (Remove Discovered Target)

To remove a nonpersistent discovered target, issue the `-dtrem` command as follows:

```
-dtrem <hba_port_inst> <target_instance>
```

To obtain the target instance, use the command [“-dtdsp \(Display Discovered Targets\)” on page 6-8](#).

## **-dtrema** (Remove Discovered Target)

To remove all nonpersistent discovered targets, issue the `-dtrema` command as follows:

```
-dtrema <hba_port_inst>
```

## **-dumpcore** (Retrieve FW Coredump Record)

To dump the RAM memory to a file name of your choice, issue the `-dumpcore` command as follows:

```
-dumpcore <hba_port_inst> [File name]
```

## **-dumpnvram** (Retrieve FW Flash & NVRAM Record)

To dump the NVRAM to a file name of your choice, issue the `-dumpnvram` command as follows:

```
-dumpnvram <hba_port_inst> [File name]
```

## **-dv**

To see the version of the iSCSI adapter driver(s) installed, issue the `-dv` command as follows.

```
-dv
```

## **-edchap** (Edit a CHAP Entry)

To change the value of the specified CHAP entry, issue the `-edchap` command as follows:

```
-edchap <hba_port_inst> <CHAP> <Parameter> | <Parameter Alias>
<Value> [<Parameter> | <Parameter Alias> <Value> ...]
```

You can repeat the *<Parameter>|<Parameter Alias> <Value>* parameter pair to change multiple values as shown in [Table 6-2](#). Press ENTER after each entry.

**Table 6-2. CHAP Parameters**

| Parameter  | Parameter Alias | Value            |
|------------|-----------------|------------------|
| CHAPName   | CNAME           | character string |
| CHAPSecret | CSECRET         | character string |
| CHAPBidi   | CBIDI           | on or off        |

Issuing this command resets the adapter.

## -ei

To get information about errors and exit codes for noninteractive mode, issue the -ei command as follows (no parameters are required):

```
-ei
```

## -f

### (Update Firmware Image—Specific HBA)

The firmware runs the tasks involved in data management, the iSCSI protocol, and general adapter functioning. Use the -f command to update the firmware image.

To update the adapter firmware from a specified file, issue the -f command as follows:

```
-f <hba_port_inst> <file name>
```

Both parameters are required. The *<hba\_port\_inst>* is the adapter to update. The *<file name>* is the file containing the firmware image to upload to the adapter. Be sure to use full path names. Issuing this command resets the adapter.

## -g

### (Display General System Information)

To view general system information, issue the -g command as follows:

```
-g
```

The QConvergeConsole CLI shows the following:

- Host name
- Host version

- Host type
- User type

## **-gcr** (Retrieve FW Crash Record)

To get crash record information for the specified adapter port, issue the `-gcr` command as follows:

```
-gcr <hba_port_inst> <Crash Output File>
```

## **-h** (Help)

To view the QConvergeConsole CLI help file, issue the `-h` command as follows:

```
-h
```

For convenience, you can send the output to a file for easier viewing or printing as follows:

```
gauccli -pr iscsi -h >file.txt
```

## **-i** (List All QLogic iSCSI HBA Ports Detected)

Use the `-i` command to view a list of all detected adapter port instances ("ports") in the system. Each port instance (*hba\_port\_inst*) is identified by its number.

To view general adapter information, issue the `-i` command as follows:

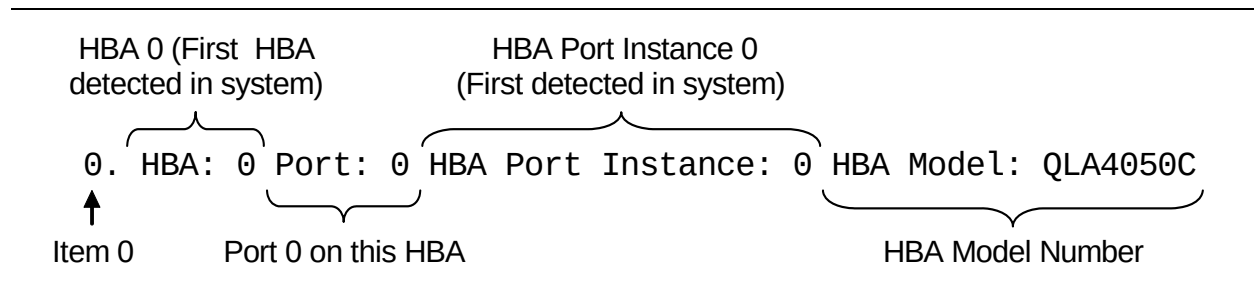
```
-i [hba_port_inst]
```

When you use *hba\_port\_inst*, information for only that adapter port instance is shown. When you do not use *hba\_port\_inst*, information for all ports in the system is shown.

The following adapter (port) information is shown:

- Item number (order in which ports were detected)
- Adapter number, that is, the number of this adapter in the detection process
- Adapter port number (0 or 1) for this adapter
- Port instance number in the system (two-port adapters appear twice)
- Port Instance on the adapter
- Adapter model number (for example, QLA4010)
- Adapter serial number (for example, TFC0105C51704)
- Firmware version (for example, 3.0.1.45)
- Connection medium
- IP address (IP address of the adapter port instance)
- iSCSI name (adapter port iSCSI name; the QLogic default name or one you assign)
- Alias (adapter port instance iSCSI alias name that you assign)

Figure 6-1 shows how to read the display.



**Figure 6-1. Adapter Port Information**

The following shows an example of the adapter port information:

```
0. HBA 0 Port: 0 HBA Port Instance: 0 HBA Model: QLA4050C
HBA Serial Number: FS20525B03135FW Version: 2.2.4.45 Type: Copper
IP Address: 192.168.3.7
Alias:
iSCSI name: iqn.2000-04.com.qlogic.fs10506a02810.1

1. HBA 1 Port: 0 HBA Port Instance: 1 HBA Model: QLA4062C
HBA Serial Number: AS40637A04673FW Version: 3.0.1.18 Type: Copper
IP Address: 192.168.3.22
Alias:
iSCSI name: iqn.2000-04.com.qlogic.as10506a02810.1

2. HBA 1 Port: 1 HBA Port Instance: 2 HBA Model: QLA4062C
HBA Serial Number: AS40637A04673FW Version: 3.0.1.18 Type: Copper
IP Address: 192.168.3.38
Alias:
iSCSI name: iqn.2000-04.com.qlogic.as10506a02810.1

3. HBA 2 Port: 0 HBA Port Instance: 3 HBA Model: QLA4010
HBA Serial Number: FS20407B03135FW Version: 3.0.1.17 Type: Copper
IP Address: 192.168.3.72
Alias:
iSCSI name: iqn.1991-05.com.microsoft.steinway.qlogic.org
```

## **-import** (Import HBA Configuration)

To import an adapter configuration from a file (see “[-save \(Save Host Configuration\)](#)” on page 6-24), issue the `-import` command as follows:



```
-import <hba_no_target> <hba_no_Source> [A] [F N I T C B]
<file_name>
```

Where:

*<hba\_port\_inst>* = The specific adapter to which the data is written  
*<hba\_port\_inst\_source>* = The file with the adapter configuration you want.  
 A = Configure all imported parameters (F, N, I, T, C, B).  
 F = Configure the firmware.  
 N = Configure the network parameters.  
 I = Configure the iSCSI name.  
 T = Configure the targets.  
 C = Configure the CHAP table.  
 B = Configure the boot parameters (only valid when the T [targets] parameter is also specified).  
*file\_name* = The XML file with the adapter configuration. See [“-save \(Save Host Configuration\)” on page 6-24](#)).

## **-ipdhcp** (Configure IP Settings)

To set the adapter to get its IP address, subnet mask, and gateway from the DHCP server, issue the -ipdhcp command as follows:

```
-ipdhcp <hba_port_inst> [-option]
```

If the adapter port had prior values set manually, the DHCP values replace them.

Where *[option]* includes:

- DISN4 = Disable IPv4 network
- IPV4DHCP = Set DHCP ON
  - IP = Set IP Address
  - IPNM = Set IP Netmask
  - IPGW = Set IP Gateway
- DISN6 = Disable IPv6 network setup
- IPLLA = Set local link address automatically
  - IPLL = Set local link address manually
  - IPRA = Set IPR0 and IPR1 automatically
  - IPR0 = Set routable address 0 manually
  - IPR1 = Set routable address 1 manually
  - IPRAA = Set router address automatically
  - IPRR = Set router address manually

## **-isns** (Configure iSNS Settings)

To configure the iSNS settings for QLA4010 and QLA405x, issue the `-isns` command as follows:

```
-isns <hba_port_inst> [iSNS IP Address] [iSNSPORT <port_number>]
```

If you do not specify the `[iSNS IP Address]` parameter, the command disables the iSNS client on the specified adapter port.

Issuing this command resets the adapter.

## **-isns6** (iSNS Settings)

To configure the iSNS settings for IPv6 (QLE406x and newer), issue the `-isns6` command as follows:

```
-isns <hba_port_inst> [iSNS IPv6-or-IPv4 Address] [-iSNSPORT  
<port_number>]
```

If you do not specify the `[iSNS IPv6-or-IPv4 Address]` parameter, this command disables the iSNS client on the specified adapter port.

Issuing this command resets the adapter.

## **-l** (List LUN Information)

Use the `-l` command to view the following information about LUNs attached to the selected target:

- Adapter/Target/LUN number
- Vendor name
- Product ID
- Product revision
- LUN size

You can view this information for a specific LUN or for all LUNs attached to the selected target.

To list the LUN information, issue the `-l` command as follows:

```
-l <hba_port_inst> <Target ID [LUN ID]>
```

If you do not include the `LUN ID` parameter, all the LUNs on the specified target are shown.

## **-lcd** (Display Configured Link Configuration)

To display the configured link settings, issue the `-lcd` command. Note that because a system reset is required for changes in link configuration to take effect, the settings shown here may be different than the operating link configuration.

## **-linkchap** (Assign a CHAP Entry to a Target)

To link a CHAP entry to a target, issue the `-linkchap` command as follows:

```
-linkchap <hba port#> <chap#> <targetID> [<TGTBCA> <on|off>]
```

Where `TGTBCA` is an optional parameter to turn BIDI CHAP authentication on or off for this target connection.

For example, to link a CHAP entry 9 on port 2 to target 10 with BIDI CHAP turned on, run the following command:

```
-linkchap 2 9 10 TGTBCA on
```

This command does *not* reset the adapter.

## **-lp**

The IPv6 local prefix list is created based on prefix information from router advertisements. Destination IP addresses containing these prefixes can be reached directly, without going through an intermediate router.

To display the local prefix list, issue the `-lp` command as follows:

```
-lp <hba_port_inst>
```

## **-model**

To view the adapter “model(s)” (board type[s]) in the system, issue the `-model` command as follows:

```
-model
```

## **-n** (Configure iSCSI Settings)

To set to adapter iSCSI parameters for single-port adapters, issue the `-n` command as follows:

```
-n <HBA No.> <Config Name|Config Alias> <Value>  
[<Config Name|Config Alias> <Value> ...]
```

After you issue this command, the adapter is reset.

You can repeat the `<Config Name|Config Alias> <Value>` parameter set for as many parameters as you want to configure. [Table 6-3](#) lists these parameters and their ranges of values.

**Table 6-3. Adapter Parameters**

| Configuration Name              | Configuration Alias | Value             |
|---------------------------------|---------------------|-------------------|
| <b>Add Firmware Options</b>     |                     |                   |
| AFW_AutoConnect                 | AFWAC               | on or off         |
| AFW_Device_Timeout              | AFWDT               | on or off         |
| AFW_Delayed_Ack                 | AFWDACK             | on or off         |
| ExeThrottle                     | ET                  | 0–32767           |
| FirstBurstLen                   | FB                  | 0–32767           |
| <b>IP Options</b>               |                     |                   |
| IP_ARP_Redirect                 | IPARP               | on or off         |
| IP_Address                      | IPAD                | IP address format |
| IP_Subnet_Mask                  | IPSM                | IP address format |
| IP_Gateway                      | IPGW                | IP address format |
| <b>iSCSI Options</b>            |                     |                   |
| Force_Negotiate_Main_iSCSI_Keys |                     |                   |
| iSCSI_Alias                     | IALS                | on or off         |
| iSCSI_Data_Digests              | IDD                 | on or off         |
| iSCSI_Discovery_Logout          | ID                  | on or off         |
| iSCSI_Header_Digests            | IHD                 | on or off         |
| iSCSI_Immediate_Data            | IID                 | on or off         |
| iSCSI_Initial_R2T               | IIR2T               | on or off         |
| iSCSI_Snack                     | ISNACK              | on or off         |
| iSCSI_Strict_Login              | IS                  | on or off         |

**Table 6-3. Adapter Parameters (Continued)**

| Configuration Name           | Configuration Alias | Value                  |
|------------------------------|---------------------|------------------------|
| <b>TCP Options</b>           |                     |                        |
| TCP_Auto_Discovery           | SLPADIS             | on or off              |
| TCP_DHCP                     | TCPDHCP             | on or off              |
| TCP_Nagle                    | TCPN                | on or off              |
| TCPMaxWindowSize             | TCPMWS              | 1000h–FFFFh            |
| TCP_Time_Stamp               | TCPTMS              | on or off              |
| <b>Miscellaneous Options</b> |                     |                        |
| KeepAliveTO                  | KATO                | 0–65535                |
| Large_Frames                 | LRGFRM              | on or off <sup>a</sup> |
| MaxBurstLen                  | MBL                 | 0–65535                |
| MaxOutstandingR2T            | MOR2T               | 0–65535                |

<sup>a</sup> Not for QLA4010

## **-nc** (Display Neighbor Cache)

To display the neighbor cache (IPv6 only), issue the -nc command as follows:

```
-nc <hba_port_inst>
```

## **-netconf** (Configure IPv6 Settings)

To configure the network settings, issue the -netconf command as follows:

```
-netconf <hba_port_inst> [-IP <IP Address>] [-IPNM <Subnet Mask>]  
[-IPGW <Gateway IP Address>]
```

Where:

- IP <IP Address> = Set the IP address of the adapter
- IPNM <Subnet Mask> = Set the subnet mask of the adapter
- IPGW <Gateway Address> = Set the gateway
- DISN4 = Disable IPv4 network setup
- IPv4DHCP = Set DHCP on

Setting the IP address, subnet mask, or gateway turns off DHCP address resolution for the IP configuration, if it is on.

## **-netconf6** **(Configure IPv6 Settings)**

To manually configure the IPv6 network settings, issue the -netconf6 command as follows:

```
-netconf6 <hba_port_inst> [-IP <IPv6 Address>] [-IPNM <Subnet  
Mask> <IPv6>] [-IPGW <Gateway Address>]
```

Where [*option*] includes the following:

- IP <IP Address> = Set the IP address of the adapter
- IPNM <Subnet Mask> = Set the subnet mask of the adapter
- IPGW <Gateway Address> = Set the gateway
- DISN4 = Disable IPv4 network setup<sup>a</sup>
- IPV4DHCP = Set DHCP on<sup>b</sup>
  - IP = Set IP address
  - IPNM = Set IP netmask
  - IPGW = Set IP gateway
- DISN6 = Disable IPv6 network setup<sup>c</sup>
- IPLLA = Set local link address automatically<sup>d</sup>
  - IPLL = Set local link address manually<sup>d</sup>
- IPR01A = Set IPR0 and IPR1 automatically<sup>e</sup>
  - IPR0 = Set routable address 0 manually<sup>e</sup>
  - IPR1 = Set routable address 1 manually<sup>e</sup>
- IPRRA = Set router address automatically<sup>f</sup>
  - IPRR = Set router address manually<sup>f</sup>

<sup>a</sup> Mutually exclusive with the other IPv4 commands

<sup>b</sup> Mutually exclusive with -IP/-IPNM/IPGW

<sup>c</sup> Mutually exclusive with -IPLLA/-IPLL

<sup>d</sup> Manual or automatic—mutually exclusive

<sup>e</sup> Manual or automatic—mutually exclusive

<sup>f</sup> Manual or automatic—mutually exclusive

Setting the IP address, subnet mask, or gateway turns off DHCP address resolution for the IP configuration, if it is on. Some examples of usage follow.

To set only the local link address, issue the following:

```
-netconf6 0 -IPLL fe80::1
```

To set everything automatically, issue the following:

```
-netconf6 0 -IPLLA -IPR01A -IPRRA
```

To disable the IPv4 network, issue the following:

```
-netconf6 0 -DISN4
```

To disable the IPv4 and IPv6 networks, issue the following:

```
-netconf6 0 -DISN4 DISN6
```

## **-nh** (Configure iSCSI Settings)

To set the adapter-level parameters for multiport adapters, issue the `-nh` command as follows:

```
-nh <hba_port_inst> <Config name | Config alias> <Value>  
[<<Config name | Config alias>> <Value> ...]
```

The valid parameters and their values are shown in [Table 6-3](#).

## **-pa** (Add a Target)

To add persistent targets, issue the `-pa` command as follows:

```
-pa <hba_port_inst> <IP Address> [-PORT port number] [-INAME iSCSI  
Name]
```

If you do not specify a port number `[-PORT port number]`, the QConvergeConsole CLI uses the default number 3260. If you do not specify an iSCSI name `[-INAME iSCSI Name]`, the QConvergeConsole CLI uses the default value, an empty string.

## **-pad**

To view all targets for a port, issue the `-pad` command as follows:

```
-pad <hba_port_inst>
```

## **-pb** (Bind Target)

To bind a target (make it persistent), issue the `-pb` command as follows:

```
-pb <hba_port_inst> <Target ID>
```

## **-pbootcode** (Set Primary Boot Target Information)

To set the primary boot-code boot target and LUN, issue the -pbootcode command as follows:

```
-pbootcode <hba_port_inst> <tgt> <lun>
```

This command is not active for the PPC.

For information on secondary boot code, see [“-sbootcode \(Set Secondary Boot Target Information\)” on page 6-24](#).

## **-pdt**

To disable a target, issue the -pdt command as follows:

```
-pdt <hba_port_inst> <targetId>
```

## **-pet**

To enable a target, issue the -pet command as follows:

```
-pet <hba_port_inst> <targetId>
```

## **-pinfo** (Port Information)

To view port information for the specified adapter, issue the -pinfo command as follows:

```
-pinfo <hba_port_inst>
```



## **-ping** (Ping Target)

To ping the target device, issue the `-ping` command as follows:

```
-ping <hba_port_inst> <IP Address> <Ping Count> <Packet size>  
<IPv6 source address>
```

The following provides the *IPv6 source address* values:

- 0 = Do not care
- 1 = Local link
- 2 = Address 0
- 3 = Address 1

Where:

*hba\_port\_inst* = The adapter port that sends the ping  
*IP Address* = The IP address to ping  
*Ping Count* = The number of ping packets you want (1–[2<sup>31</sup>]).  
*Packet Size* = The size of the ping packet 0–64kB  
*IPv6 Source Address* = Origin device

## **-ps**

To view persistent targets for the adapter port, issue the `-ps` command as follows:

```
-ps <hba_port_inst> [Target ID]
```

If you do not specify the [*Target ID*], the QConvergeConsole CLI displays all targets for the specified adapter port number. If you do not specify the port instance [*hba\_port\_inst*] or the target ID, [*Target ID*], the QConvergeConsole CLI displays all target IDs for all adapter ports in the system.

## **-pu**

To unbind a persistent target, issue the `-pu` command as follows:

```
-pu <hba_port_inst> <Target ID>
```

## **-px**

To unbind all persistent targets on the adapter, issue the `-px` command as follows:

```
-px <hba_port_inst>
```

## **-r** **(Update ROM Image)**

### **CAUTION!**

- Before attempting to update the ROM image, contact QLogic Customer Support.
- Before updating the ROM image, ensure that no I/O processes are running.
- An administrator must take necessary actions to ensure changes are fully recognized by the operating system (reboot, flush cache, sync disk, and so forth).
- A ROM update and reset are performed that requires that all applications accessing impacted LUNs must be shut down.
- Any impacted systems must be unmounted.

Be sure to select **Save changes and reset HBA** afterwards; otherwise, the changes do not take effect.

Type the following command to update the specified adapter's ROM:

```
-r <hba_port_inst> <file_name>
```

The <hba\_port\_inst> parameter is the adapter instance to update. The <file\_name> parameter indicates the file containing the ROM image to upload.

## **-rdf**

To restore the specified adapter port's firmware values to the factory default values, issue the -rdf command:

```
-rdf <hba_port_inst>
```

Issuing this command resets the adapter.

## **-rdh** **(Port Restore Factory Defaults)**

### **NOTE:**

The -rdh command is not supported in the QLA4010 adapter.

Type the following command to restore the specified adapter port's settings to their factory defaults:

```
-rdh <hba_port_inst> [A] [F N I T C V]
```

Where:

*hba\_port\_inst* = Adapter port whose factory settings you want to restore.

A = Restore the factory settings for the entire adapter (firmware parameters, network parameters, iSCSI name, targets, CHAP table, NVRAM, and IPsec).

Issuing this command resets the adapter. If you use this option, no others are necessary.

F = Restore firmware parameters to the factory default values.

Issuing this command resets the adapter.

N = Restore network parameters to the factory default values.

Issuing this command resets the adapter.

I = Restore the iSCSI name to the factory default value.

Issuing this command resets the adapter.

T = Restore the targets to the factory default values.

C = Restore the CHAP table to the factory default values.

V = Restore the NVRAM to the factory default values.

If you want to restore all areas of the adapter port, either use the A parameter or use no parameter. Otherwise, specify which areas to restore. For example, the following command restores the firmware and NVRAM to the factory defaults:

```
-rdh <hba_port_inst> F N
```

## **-rwt** **(Perform Read/Write Buffer Test)**

### **CAUTION!**

Before doing a read/write buffer test, make sure that the adapter does not have any outstanding I/O operations.

The read/write buffer test writes an 8- or 16-byte pattern to the disk's buffer and reads the written buffer back. It is nondestructive to disk data.

To do a read/write buffer test, issue the -rwt command as follows:

```
-rwt <hba_port_inst> [-STOP] [-RWPAT n] [-CNT <Test Count>]
```

Where:

- STOP = Stop the test if a failure occurs.
- RWPAT *n* = Set the pattern to write to the buffer, one of the following:
  - n* = 1 (8 bytes of 0x55h) (default)
  - n* = 2 (8 bytes of 0x5Ah)
  - n* = 3 (16 bytes of 0xAAh)
  - n* = 4 (16 bytes of 0xFFh)
- hba\_port\_inst* = The adapter port tested.
- CNT = Used with *Test Count* to set the number of tests, with one test as default.
- Test Count* = The number of tests to do, with one as default. Enter a number between 2–(2<sup>31</sup>) for more than one test.

To continue, press ENTER.

## **-save** (Save Host Configuration)

To save the configuration (in XML) of all adapters detected, issue the -save command as follows:

```
-save <file_name>
```

## **-sbootcode** (Set Secondary Boot Target Information)

To set the secondary boot target and LUN, issue the -sbootcode command as follows:

```
-sbootcode <hba_port_inst> <Boot Target> <Boot LUN>
```

Issuing this command saves the configuration data to an XML file portable to all platforms supported by iSCSI. This command is not active for PPC.

For primary boot code, see [“-pbootcode \(Set Primary Boot Target Information\)” on page 6-20](#).

## **-sbootcodeid** (Set Alternative Client ID)

To set the alternative client ID (seven characters maximum), issue the -sbootcodeid command as follows:

```
-sbootcodeid <hba_port_inst> <Alternative Client ID>
```

This command is not active for the PPC.

## -sdmrc

To return the SDM library return code, issue the `-sdmrc` command as follows:

```
-sdmrc [Simple_Pattern]
```

When [*simple pattern*] is given, only the lines containing that string are printed.

## -stadd

To add a send target for discovery, issue the `-stadd` command as follows:

```
-stadd <hba_port_inst> <ip address> [chapIndex] [<TGBCA> <value>]
```

Where <value> can be **on** or **off**; **on** causes the send target login to require bidirectional CHAP.

### NOTE:

To use the `-stdsp`, `-strd`, and `-strem` commands, you must first run the `-stadd` command at least once to add a send target for discovery only (no target login). After using the `-stadd` command to add at least one send target for discovery only, then you can use the `-stdsp` command to display all send targets for discovery, along with the appropriate index to be used for the `-strd` and `-strem` commands. The `-strd` command (with the target instance from the `-stdsp` command) must be used to perform the send target discovery to discover all target portals.

After a send target for discovery only has been added by means of the `-stadd` command and discovered by the `-strd` command, or the iSNS has been configured (with a valid iSNS server), a list of targets and target portals is discovered. When you issue the `-stdsp` command, the discovered target portals are displayed. The displayed discovered target portals are not logged into targets at this time, but are local to the application and are available for login. To log a discovered target portal into a target, issue the `-dtli` command. To log all discovered target portals into targets, issue the `-dtlia` command. To remove discovered target portals from the list, issue the `-dtrem` command.

## -stat (Display Port Statistics)

To view the port statistics for the specified adapter, issue the `-stat` command as follows:

```
-stat <hba_port_inst>
```

## **-stathba**

To display adapter-level statistics, issue the `-stathba` command as follows:

```
-stathba <hba_port_inst>
```

## **-stdsp**

To view the send targets, issue the `-stdsp` command as follows:

```
-stdsp <hba_port_inst>
```

To use the `-stdsp` command, you must first run the `-stadd` command at least once to add a send target for discovery only. For more information, see [“-stadd” on page 6-25](#).

## **-strd**

To execute a rediscovery of targets for the specified send target, issue the `-strd` command as follows:

```
-strd <hba_port_inst> <target_instance>
```

To obtain the target instance, use the `-dtdsp` command.

To use the `-strd` command, you must first run the `-stadd` command at least once to add a send target for discovery only. For more information, see [“-stadd” on page 6-25](#).

## **-strem**

To remove the specified send target, issue the `-strem` command as follows:

```
-strem <hba_port_inst> <target_instance>
```

To obtain the target instance, use the `-dtdsp` command.

To use the `-strem` command, you must first run the `-stadd` command at least once to add a send target for discovery only. For more information, see [“-stadd” on page 6-25](#).

## **-strema**

To remove all send targets, issue the `-strema` command as follows:

```
-strema <hba_port_inst> <target_instance>
```

## **-t**

### **(Display Target Information)**

To view target information, issue the `-t` command as follows:

```
-t <hba_port_inst> [Target ID]
```

If you only specify the `<hba_port_inst>` parameter, the QConvergeConsole CLI displays target information for all targets on the specified adapter port. If you specify the optional parameter `[Target_ID]`, the QConvergeConsole CLI displays information on the specified target only. This includes both persistent and dynamic targets.

## **-tc** (Configure Target Parameters)

To set target parameter settings, use the `-tc` command. For a list of target parameters you can configure, issue the following:

```
-tc <hba_port_inst> <Target_ID> <config_name | config_alias>  
<value> <config_name | config_alias> <value>
```

## **-tp**

To see information stored in Flash memory about persistent targets, issue the `-tp` command. If you enter only the adapter port instance number, the QConvergeConsole CLI returns information for all targets of that adapter port instance. If you enter the optional Target ID, the QConvergeConsole CLI returns information for that target only, for example:

```
-tc <hba_port_inst> <Target_ID>
```

## **-ts**

To see summary information about both persistent and nonpersistent targets, issue the `-ts` command as follows:

```
-ts [hba_port_inst] [Target_ID]
```

## **-ver** (Display Program Version Information)

To view the QConvergeConsole CLI program version, issue the `-ver` command as follows:

```
-ver
```

## **-vpd** (Display VPD Information)

To view VPD information, if any, issue the `-vpd` command as follows:

```
-vpd <hba_port_inst>
```

Available VPD information varies by adapter manufacturer. QLogic adapters ship with VPD information shown as N/A.









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